
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is among the most widely used neighbor-embedding methods, and its empirical behavior is governed by a measurable balance between attractive and repulsive forces. We frame these forces as comparable functional shapes and use that framing to explain cluster formation, visualization quality, and differences between UMAP and related methods. The comparisons show a consistent division of labor: repulsion primarily controls cluster boundaries and inter-cluster distance, whereas attraction determines when local pairs contract, when they unexpectedly expand, and how stable the embedding remains across initializations. This quantitative view clarifies why UMAP benefits from learning-rate annealing, why newer bounded attraction shapes are less sensitive to fixed learning rates, and why modifying attraction improves consistency under random initialization. Across the reported metrics and visual comparisons, the force-shape analysis makes the practical significance of UMAP-style embeddings clearer: it preserves the original quantitative facts while making the mechanisms behind robustness, compactness, and accuracy more interpretable.

1 Introduction

High-dimensional datasets now appear in many domains where the first practical question is comparative: which points are close, which groups separate, and how stable are those patterns under visualization? Dimensionality reduction (DR) methods answer this question by converting high-dimensional relations into low-dimensional embeddings whose empirical quality can be judged through neighborhood preservation, cluster separation, and reproducibility. Two of the most used methods, t -distributed stochastic neighbor embedding (Maaten and Hinton, 2008) and uniform manifold approximation and projection (UMAP) (McInnes et al., 2018), implement this conversion through attraction-repulsion dynamics: similar points are pulled together, while dissimilar points are pushed apart. Because these algorithms are unsupervised, their value depends on how reliably they expose intrinsic structure without labels. That reliability is central in clustering (Becht et al., 2019), exploratory data analysis (Fleischer and Islam, 2020), anomaly detection in semiconductor manufacturing (Fan et al., 2021), visual search (González-Márquez et al., 2024), time series analysis (Altin and Cakir, 2024), studying representation convergence (Huh et al., 2024), and outlier image detection (Islam and Fleischer, 2024). In each setting, quantitative evidence about local preservation, label separation, and run-to-run consistency determines whether an embedding is merely visually appealing or practically interpretable.

Recent attraction-repulsion methods (Amid and Warmuth, 2019; Agrawal et al., 2021; Wang et al., 2021; Narayan et al., 2021; Yang et al., 2022; Wang et al., 2025; Kury et al., 2025) improve different aspects of the embedding, but their empirical advantages are often presented through heuristics rather than through comparable quantities. This is consequential because DR necessarily introduces distortions (Chari and Pachter, 2023); practitioners therefore need evidence that separates useful structure from artifacts. In practice, compact clusters can be produced by initialization choices, learning-rate schedules, and kernel-function tuning, but these interventions are rarely compared at the level of the attractive and repulsive forces that actually move points. The central gap is thus quantitative and mechanistic: the parameters that determine cluster compactness, inter-cluster margins, and stability are hidden inside functional forms, making it difficult to explain why one setting, method, or metric outcome is better than another.

In this paper, we decompose the forces into directly comparable attraction and repulsion shapes and use those shapes to organize the empirical evidence. The resulting comparisons identify attraction as the common driver behind three reported phenomena: the need for learning-rate annealing, the difficulty of obtaining consistent outputs from random

initialization, and the formation of new cluster structure. Repulsion, by contrast, most clearly changes inter-cluster distances and cluster compactness. Our specific contributions are:

1. We formulate attraction and repulsion shapes from the attractive and repulsive forces and derive explicit per-iteration conditions for distance contraction and expansion, giving a common comparison scale for the algorithms (Section 4).
2. We show that UMAP’s attraction shape can cause both contraction and expansion of pairwise distance; comparing this shape against other algorithms explains the empirical role of learning-rate annealing (Section 5).
3. We modify the attraction shape and quantify the resulting gain in consistency under random initialization, supporting the interpretation that UMAP encodes a reproducible structure that can be reached more reliably (Section 5.1).
4. We analyze repulsion shapes to separate their contribution to inter-cluster distance from the attraction-driven emergence of cluster structure (Section 5.2).
5. We compare attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG-*t*-SNE, making the stability margins and similarities among the methods explicit (Section 6).

We center the main text on UMAP because it is a de facto standard across application areas, and we introduce other algorithms when their shapes provide an informative baseline or contrast. Additional method-specific details appear in the appendix. The codes used in the research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor embeddings can be organized by the graph they compare and the quantitative objective they optimize. Stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its *t*-distributed extension (*t*-SNE) (Maaten and Hinton 2008) use dense pairwise graphs, normalize graph weights into probability distributions, and optimize the discrepancy between high- and low-dimensional similarities. Other methods, including locally linear embedding (Roweis and Saul 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002), instead use a *k*-nearest neighbor (*k*-NN) graph and spectral decomposition. Tang et al. (Tang et al. 2016) combined the *k*-NN graph with iterative optimization and removed normalization in the low-dimensional space. UMAP (McInnes et al. 2018) further removed normalization in both spaces, so the embedding is produced by pairwise interactions alone. In this progression, the empirical comparison reduces to two measurable forces: attraction preserves local neighborhoods, and repulsion separates dissimilar points. PaCMAP (Wang et al. 2021) and NEG-*t*-SNE (Damrich et al. 2023) build on these same ingredients. For a recent survey of methods, see de Bodt et al. (2025).

Theoretical work has increasingly made these empirical comparisons more precise. Early analysis of SNE showed that well-clustered input data remain well-clustered in the embedding space (Shaham and Steinerberger 2017), and analysis of *t*-SNE showed that the number of embedded clusters lower-bounds the number of original clusters (Linderman and Steinerberger 2019), with subsequent refinements in (Arora et al. 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Because *t*-SNE and UMAP share a common neighbor-embedding framework yet often produce visually different outputs, much recent work has quantified their connection (Böhm et al. 2022; Damrich et al. 2023; Draganov et al. 2023). Böhm et al. (2022) placed Laplacian Eigenmaps, ForceAtlas2 (Jacomy et al. 2014), UMAP, and *t*-SNE on a common spectrum; LE and *t*-SNE are connected through the early exaggeration phase (Cai and Ma 2022); and UMAP and *t*-SNE can be related through contrastive estimation (Damrich et al. 2023). Around the same time, Hu et al. (2023) independently linked contrastive learning and SNE. Other recent explanations use probabilistic formulations (Ravuri et al. 2023; Ravuri and Lawrence 2024), kernel analyses (Draganov and Dohn 2023), and information geometry (Kolpakov and Rocke 2024). Our contribution complements these accounts by foregrounding the numerical margins in the attraction and repulsion shapes that predict observed embedding behavior.

3 Uniform Manifold Approximation and Projection

UMAP begins with a high-dimensional graph of the dataset $X = \{x_i \in \mathbb{R}^n | i = 1, \dots, N\}$, where each edge weight is a pairwise score $p_{i,j} = f_h(d(x_i, x_j))$, typically in $[0, 1]$ (for details, see Appendix H). This graph supplies the quantitative baseline against which the low-dimensional embedding is optimized.

The graph of the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is given by a differentiable function

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where the parameters a and b determine the density and tail behavior of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d regulates the distance between the two nearest low-dimensional points. UMAP aims to minimize the following cross-entropy loss function:

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term contributes the attractive update and the second term contributes the repulsive update, so the loss already separates the two empirical effects we compare throughout the paper. Rather than optimizing every pair at each iteration, UMAP uses negative sampling (Mikolov et al. 2013, Tang et al. 2016): for each positive edge with $p_{ij} > 0$, several randomly selected negative edges provide the repulsive comparison. The attractive force is applied on the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive force is applied on the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t is the step number. The negative-edge update changes y_i but not y_j , a detail that matters when comparing the effective strength of attractive and repulsive steps. For a detailed analysis of UMAP's loss, see Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The updates (4,6) become easier to compare when each gradient $\nabla_{y_i^t} \log(q_{ij})$ ($\nabla_{y_i^t} \log(1 - q_{ij})$) is decomposed into two factors: a scalar coefficient determined by the current distance $\|y_i - y_j\|_2$ and the displacement vector $(y_i - y_j)$. We call this scalar coefficient the attraction (repulsion) shape. Although UMAP is the running example, the same shape-based comparison applies to any method whose optimization is driven by attraction and repulsion.

For a positive edge, writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, exposes the attraction shape directly in the update:

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ is the attraction shape, and the Euclidean metric gives $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. For a negative edge, the analogous decomposition isolates the repulsion shape:

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape. These two scalar functions give a common numerical language for comparing methods whose full losses may look different.

Related decompositions appear in (McInnes et al. 2018, Agrawal et al. 2021, Draganov and Dohn 2023), but they were used for different quantitative purposes. The original UMAP paper (McInnes et al. 2018) used the derivative form for efficient computation, and Draganov and Dohn (2023) used it to compare algorithms from a kernel perspective. In both cases, the derivative was the object of interest. Here, the scalar functions themselves are the objects of comparison: their ranges, bounds, and distances from critical thresholds explain empirical behavior. A related formulation in Agrawal et al. (2021) used a scalar coefficient and the unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to emphasize force magnitude. Our shape decomposition keeps the distance dependence visible, which is essential for interpreting margins such as the boundary between contraction and expansion.

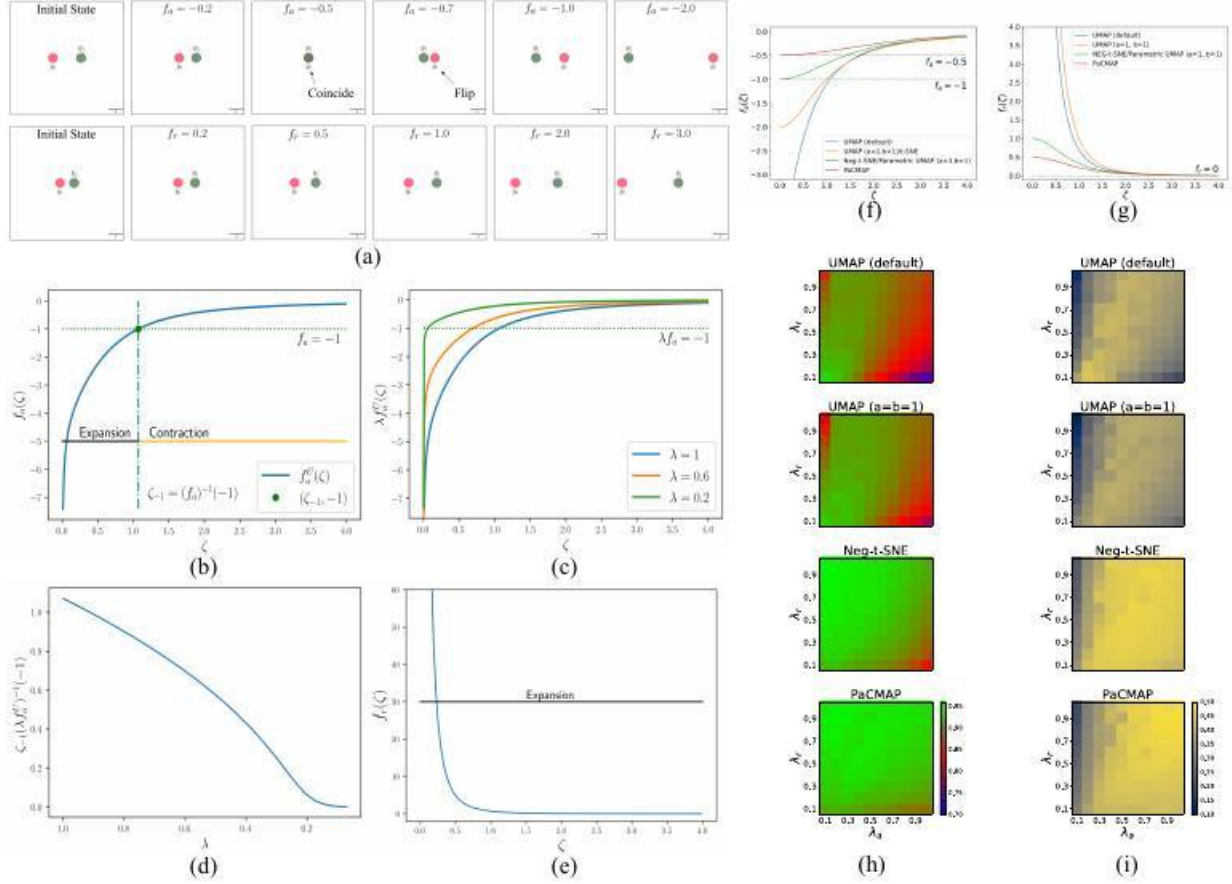


Figure 1: Quantitative comparison of attraction and repulsion shapes. (a) Pairwise effect of representative f_a values (top) and f_r values (bottom). (b) UMAP attraction shape, highlighting the range where attraction can become expansion. (c) Effective attraction shape (λf_a) across learning rates λ . (d) Minimum contraction distance (ζ_{-1}) decreases as λ decreases. (e) UMAP repulsion shape. (f) Attraction and (g) repulsion shapes for multiple embedding methods on the same scale. (h) Trustworthiness and (i) Silhouette Score as attraction learning rate (λ_a) and repulsion learning rate (λ_r) vary independently. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

4.1 Conditions for Attraction and Repulsion

We next derive per-step conditions that turn the shapes into quantitative certificates for attraction and repulsion. From the update equations (7)-(10), the following proposition gives the exact margin for contraction of a positive-edge pair y_i^t and y_j^t :

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Thus, λf_a is the effective attraction shape, and the interval $(-1, 0)$ is the stability margin for one-step contraction. In many comparisons, the unscaled f_a already indicates whether a method is likely to stay inside that margin. For a negative edge, the corresponding expansion condition is simpler:

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Adding a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ to Eq. (10) leaves this expansion conclusion unchanged. These propositions should be read as per-iteration guarantees, not as a restatement of learning-rate decay: suitable shapes can satisfy the contraction and expansion margins without relying on annealing.

Figure 1(a) makes the thresholds explicit for a pair of points. Any $f_r > 0$ increases distance, so the repulsion comparison is monotone. Attraction is more informative: $f_a \in (-1.0, 0)$ decreases distance, $f_a = -0.5$ gives the

strongest contraction to coincident points in one step, and $f_a < -1.0$ reverses the intended effect by increasing distance. These are local per-pair margins, but their repeated accumulation determines the global cluster structure evaluated in the embeddings.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition together with the low-dimensional affinity in (1), UMAP’s attraction and repulsion shapes can be compared explicitly. The attraction shape is

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. We use these two functions to interpret the default UMAP setting, whose behavior is controlled mainly by a and b , and then compare the effect of perturbing those parameters. The discussion is generally valid for $b \leq 1$, where the attraction shape is strictly increasing and therefore invertible (For derivation, see Appendix A).

Figure 1(b) plots the default UMAP attraction shape ($a = 1.58, b = 0.89$). Its magnitude becomes unbounded as $\zeta \rightarrow 0$, so the effective attraction λf_a^U eventually crosses the -1 contraction threshold from Proposition 4.1. Because the shape is invertible, the crossing defines a measurable margin, $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, which is the minimum distance at which attractive updates still contract. Distances $\zeta > \zeta_{-1}$ contract, whereas $0 < \zeta < \zeta_{-1}$ expand, showing quantitatively why an update labeled “attractive” can push very close neighbors apart.

A larger ζ_{-1} leaves a wider interval in which neighboring points can oscillate between contraction and expansion, producing visibly fuzzier boundaries. Sharp boundaries therefore correspond to the practical limit $\zeta_{-1} \rightarrow 0$, which UMAP approaches by annealing the learning rate toward zero (Figs. 1(c,d)). The repulsion shape in Eq. 12 has a different profile: it is always positive and therefore satisfies Proposition 4.2, approaches 0 as $\zeta \rightarrow \infty$, and approaches ∞ as $\zeta \rightarrow 0$ (Fig. 1(e)).

The cross-method comparison in Fig. 1(f) shows that UMAP and t-SNE are the methods whose attraction shapes enter the $f_a < -1$ regime (and hence have $\zeta_{-1} > 0$ at $\lambda = 1$). t-SNE controls this by weighting updates with the corresponding p_{ij} values, whereas UMAP controls it through learning-rate annealing. In contrast, NEG-t-SNE and PaCMAP keep f_a within $[-1, 0]$, so Proposition 4.1 holds for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$; PaCMAP’s weighted f_a (< 0.5 always) also avoids sign flips during attraction. The repulsion comparison in Fig. 1(g) shows a complementary pattern: UMAP alone has very large repulsion at small distances (unbounded in the shape, though often clipped during optimization). For additional details of these and related methods, see Appendix G.

We evaluate these shape differences on MNIST by assigning separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and sweeping both values (Figs. 1(h,i)). For UMAP, the favorable region is concentrated at lower attraction learning rates ($\lambda_a < 0.5$, with $\zeta_{-1} \simeq 0$), where both trustworthiness (T) (Venna and Kaski [2001], measuring neighborhood preservation, and silhouette score (SIL) (Rousseeuw [1987], measuring label separation, improve relative to larger λ_a . By comparison, the full tested range of $\lambda_r \in [0, 1]$ remains usable. For NEG-t-SNE and PaCMAP, where $\zeta_{-1} = 0$ by construction, the effective region spans the whole parameter range. The metric pattern therefore matches the shape analysis: UMAP must drive ζ_{-1} close to 0 for satisfactory embedding and clustering, which it does through annealing, whereas newer algorithms begin with that margin already at 0. For additional discussions, results, and illustrations of these methods on different datasets, see Section 6 and Appendices B, G and I.

5.1 Improving UMAP’s Consistency under Random Initialization

UMAP’s run-to-run consistency is strongly tied to initialization (Kobak and Linderman [2021], Wang et al. [2021]). PCA or spectral initialization supplies a structured starting point and usually yields consistent mappings despite stochastic optimization. Random Gaussian initialization is a harder evaluation condition: clusters may fail to form or may appear in different orientations across runs. The shape analysis explains this failure mode quantitatively. If two high-dimensional nearest neighbors start far apart in the embedding, UMAP’s attraction becomes too weak to close the gap; this near-sightedness (Wang et al. [2021]) is visible in the tail of the attraction shape, where $|f_a^U|$ diminishes toward zero as distance grows (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)| = 0$).

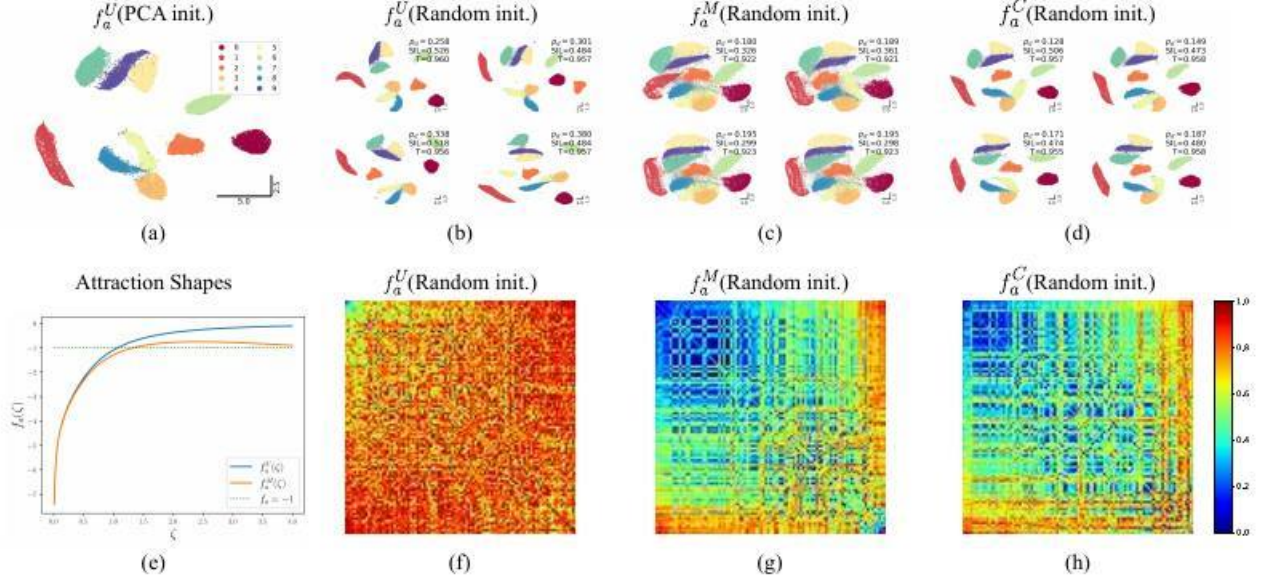


Figure 2: Random-initialization consistency on MNIST under three attraction shapes. (a) PCA-initialized reference mapping. (b-d) Four random-initialized mappings with the lowest Procrustes distance (p_d) to (a) for (b) UMAP, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices over 100 runs for (f) UMAP (0.78 ± 0.13), (g) modified (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20) shapes. Diagonal (i, i) entries are sorted by p_d from (a), and off-diagonal (i, j) entries report p_d between the i^{th} and j^{th} mappings. The lower mean $p_d \pm \text{std}$ values for modified and composite attraction quantify the improved self-similarity relative to default UMAP.

To test whether stronger long-range attraction improves this difficult condition, we induce “far-sightedness” by increasing the attraction assigned to large distances. The modified UMAP attraction shape is

$$f_a^M = f_a^U - \beta \zeta, \quad (13)$$

where β controls the added long-range attraction (we used $\beta = 0.2$, Fig. 2(e)). In loss-function terms, this shape change adds a regularizer to the attractive term (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). The modification both pulls distant neighbor pairs together and encourages early intermixing, which improves convergence from random initialization in a role analogous to early exaggeration in t -SNE. Equation 13 uses the simplest linear correction; other distance-increasing functions, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), may also be suitable.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape uses the stronger attraction only during the early epochs and then returns to the original UMAP shape, so any long-range correction is limited to the phase where random initialization is most problematic. We next compare the default, modified, and composite shapes under the same random-initialization protocol.

For the MNIST consistency experiment, we first compute a PCA-initialized reference embedding (Fig. 2(a)). We then run each attraction shape from Gaussian random initialization 100 times. Procrustes analysis (Gower [1975]) aligns each output to the reference under scaling, translation, rotation, and reflection (for details, see Appendix D.1), and Procrustes distance (p_d) provides the comparison metric. Under default UMAP, the four lowest- p_d random runs retain recognizable cluster shapes but differ in placement (Fig. 2(b)). The modified and composite shapes improve this placement consistency under the same evaluation condition (Figs. 2(c) and (d)).

We further summarize all pairwise run comparisons with a Procrustes matrix: diagonal entries are sorted by p_d from the PCA-initialized reference, and off-diagonal entries are p_d between pairs of randomly initialized mappings (for details, see Appendix D.1). This matrix plays the role of a run-similarity matrix (Foote [1999]). Default UMAP yields weak self-similarity across runs (Fig. 2(f)), with mean $p_d \pm \text{std}$ of 0.78 ± 0.13 . The modified and composite shapes reduce the mean pairwise distance to 0.49 ± 0.21 and 0.50 ± 0.20 , respectively (Figs. 2(g,h)). These margins show that UMAP encodes a reproducible structure, but default long-distance attraction can be too small to reach that structure consistently from random starts.

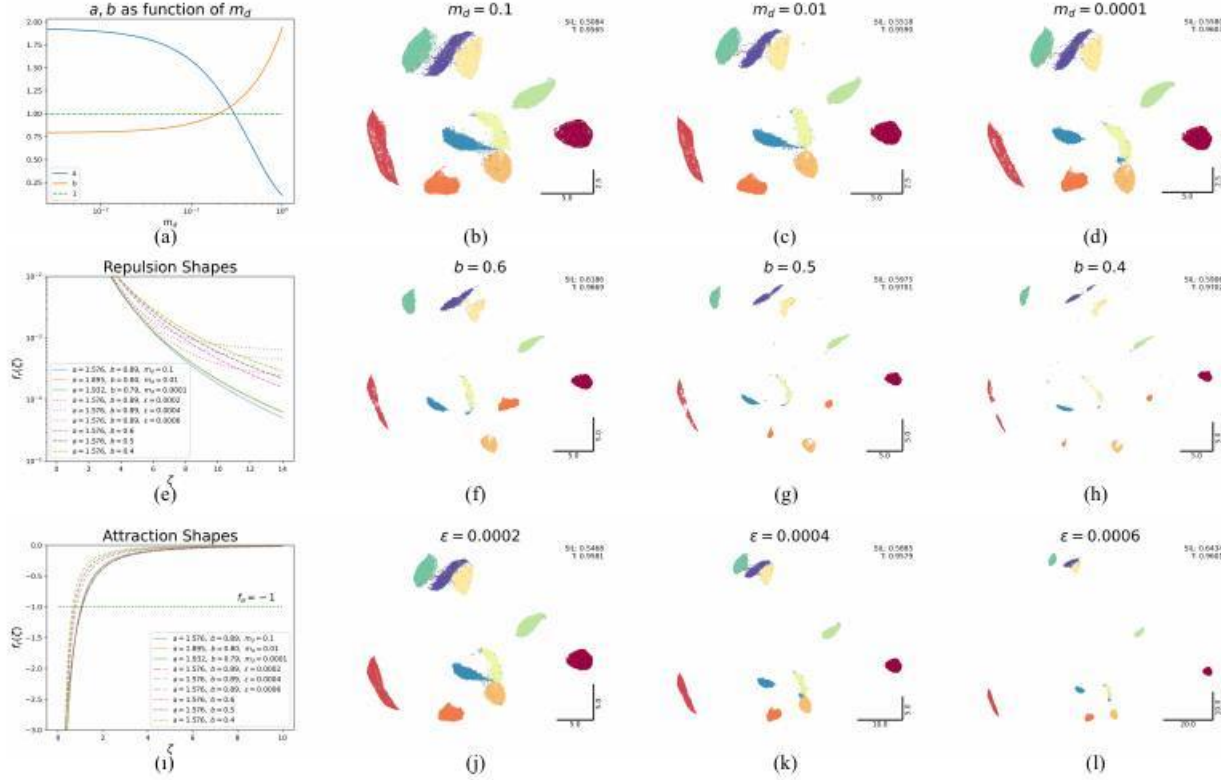


Figure 3: MNIST comparison for controlling cluster compactness and inter-cluster margins. (a) Fitting a, b through the low-dimensional distance m_d explores only a restricted parameter path. (b-d) Setting m_d to 0.1, 0.01, and 0.0001 yields limited additional compactness. (e) Repulsion shapes under alternative parameter choices. (f-h) Explicitly lowering b increases repulsion, makes clusters more compact, and reveals clusters absent under the default setting. (i) Corresponding attraction shapes. (j-l) Adding a small positive ϵ to the repulsion shape increases inter-cluster distance while leaving attraction fixed.

5.2 Cluster Formation and Compactness

UMAP users commonly control cluster formation through the minimum-distance parameter m_d in Eq. 2. The evidence in Fig. 3(a), however, shows that varying m_d explores only a restricted path through the underlying a, b values. Consequently, reducing m_d yields limited practical gain in compactness for the MNIST embeddings (Figs. 3(b-d)), because it changes the repulsion shape only modestly (Fig. 3(e)). Directly varying a and b gives a clearer comparison. Decreasing a increases repulsion, but it weakens attraction even faster and worsens near-sightedness. Decreasing b gives better control over both shapes (Figs. 3(e,i)): setting b to 0.6, 0.5, and 0.4 progressively increases inter-cluster distance and splits previously merged clusters (Figs. 3(f-h)). This pattern is consistent with the increasingly heavy-tailed kernel as b decreases, since heavy-tailed kernels produce smaller and more distinct clusters (Van der Maaten and Hinton [2008]; Kobak et al. [2019] (see Kobak et al. [2019] for how varying b in UMAP modulates the tail)). The setting separates all ten MNIST labels into distinct clusters, but changing b affects attraction and repulsion together, so their individual contributions require a controlled comparison.

To isolate repulsion, we keep attraction fixed and change only the repulsion shape by adding a small positive value (ϵ):

$$f_r^M = f_r^U + \epsilon, \quad (15)$$

with a and b held fixed. This is equivalent to adding a regularizer to the repulsive term of the loss (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \epsilon/2 \|y_i - y_j\|_2^2$), and it produces stronger repulsion on the same attraction baseline (Fig. 3(e)). As ϵ increases, inter-cluster distances increase as well (Figs. 3(j-l)). The resulting label separation resembles the effect of varying m_d : labels become more loosely separated, but no comparable new cluster structure appears. Because ϵ leaves attraction unchanged, and because varying m_d traces similar attraction shapes (Fig. 3(i)), the comparison supports the conclusion that attraction is the dominant driver of cluster formation while repulsion adjusts margins and compactness.

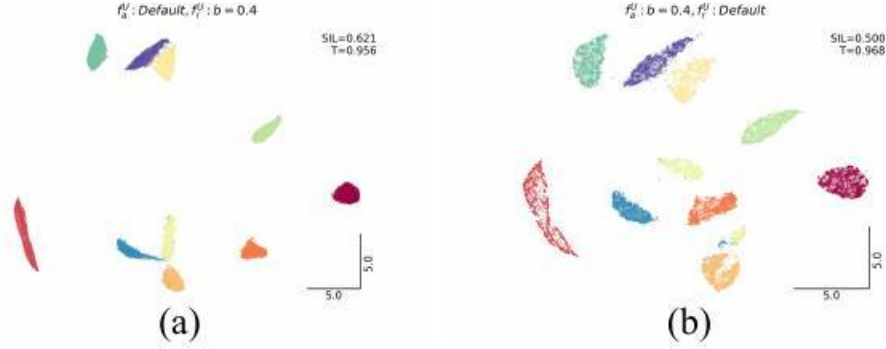


Figure 4: Controlled MNIST comparison separating attraction from repulsion. (a) Default attraction with the $b = 0.4$ repulsion shape preserves the default UMAP cluster identities but makes them more compact. (b) Default repulsion with the $b = 0.4$ attraction shape creates additional within-cluster structure and new clusters.

We therefore perform a second controlled comparison by changing only one shape at a time while setting b to 0.4 for the altered shape (Fig. 4). When attraction remains at its default and only repulsion uses the $b = 0.4$ profile, the embedding keeps the default UMAP cluster identities but makes them smaller and more compact (Fig. 4(a)). When repulsion remains at its default and only attraction uses the $b = 0.4$ profile, additional within-cluster structures appear and some previous clusters split into smaller groups (Fig. 4(b)). This side-by-side comparison assigns the empirical roles more clearly: attraction creates or refines cluster structure, while repulsion changes compactness and inter-cluster margins. Appendix G.3 shows that LocalMAP embeddings also exploit this attraction-repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

A common stabilization choice in dimensionality reduction is to set $a = 1$ and $b = 1$, which often bounds gradients and simplifies optimization. Damrich et al. (2023) examined this setting for UMAP and proposed NEG- t -SNE, which combines stable gradients with more compact clusters even when the learning rate is held constant at 1. Parametric UMAP (Sainburg et al., 2021) followed a related path: it originally used the UMAP loss, but later it adopted a numerically stable modified cross-entropy loss (Shi et al., 2023) with a logsigmoid kernel. Comparing the resulting formulas shows that the methods reach the same objective from different directions:

Proposition 6.1. *NEG- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

For UMAP with $a = 1$ and $b = 1$, the attraction and repulsion shapes are $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. This bounds attraction in $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), but leaves repulsion effectively unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). The attraction bound is not sufficient by itself: because $f_a^U < -1$ for small ζ , Proposition 4.1 and Section 5 predict that even this unity case still benefits from learning-rate annealing.

Damrich et al. (2023) analyzed UMAP’s negative-sampling loss through contrastive embedding (CE) and identified its effective kernel as $1/\zeta^2$. In the CE framework, NEG- t -SNE replaces that kernel with $1/(1 + \zeta^2)$; translated back into UMAP notation, the low-dimensional affinity becomes $q_{ij}^N = 1/(2 + \zeta^2)$. The resulting shapes are $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The comparison is favorable for stability: attraction is bounded within $[-1, 0]$, so Proposition 4.1 holds for every $\lambda \in [0, 1]$ and avoids attraction-induced oscillation. NEG- t -SNE is therefore less sensitive to annealing, and its clusters remain less fuzzy even at constant $\lambda = 1.0$. Its repulsion is also bounded within $[0, 1]$ and does not diverge as $\zeta \rightarrow 0$, reducing the chance that randomly sampled negative updates eject points from clusters. Whereas Damrich et al. (2023) emphasized bounded repulsion as the explanation for the disparity, Fig. 1 shows that the attraction margin is equally important for stability and compactness. For additional discussion on NEG- t -SNE with illustration, see Appendix F. Together with Fig. 4 this suggests a practical design space in which attraction and repulsion shapes from different methods can be mixed, such as combining UMAP attraction with NEG- t -SNE repulsion. Appendix J gives examples using UMAP, NEG- t -SNE, and PaCMAP.

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7 Discussion and Conclusion

This work reframes cluster formation in neighbor embeddings as a quantitative comparison of attractive and repulsive force shapes. The usual intuition—attraction pulls similar points together and repulsion pushes dissimilar points apart—does not specify the margins, thresholds, or evaluation conditions under which those forces succeed. By analyzing those quantities directly, we explain the empirical patterns behind compactness, fuzziness, random-initialization consistency, and learning-rate sensitivity. The main text emphasizes UMAP and NEG- t -SNE, but the same shape-based evidence applies broadly to dimensionality reduction methods, including the additional algorithms discussed in Appendix G.

Treating attraction and repulsion coefficients as functional mappings made the central numerical threshold explicit. Propositions 4.1 and 4.2 show when a single update contracts or expands distance, and Fig. 1(b) shows that UMAP’s attraction can cross into the expansion regime at short distances. Thus, UMAP’s learning-rate schedule is not merely a tuning heuristic; it reduces the contraction threshold ζ_{-1} and mitigates attraction-induced expansion.

The random-initialization experiments further show the practical significance of the shape view. Similar high-dimensional neighbors that start far apart receive weak default attraction and may not contract sufficiently. Adding a long-range term to the attraction shape directly targets this failure mode and reduces Procrustes distances across 100 random runs on MNIST, with analogous improvements on the additional datasets. These consistency gains support the interpretation that UMAP encodes a reproducible structure, and that modified attraction helps reach it more reliably from difficult starts.

The repulsion experiments assign a different empirical role to repulsion. Increasing repulsion enlarges inter-cluster distance and changes compactness; adding a small positive value to the repulsion shape achieves stronger repulsion than the standard m_d path provides, while preserving the attraction shape for a controlled comparison.

The comparison with NEG- t -SNE clarifies why bounded shapes matter. At $a = 1$ and $b = 1$, NEG- t -SNE has attraction bounded in $[-1, 0]$ and repulsion bounded in $[0, 1]$, so the contraction and expansion conditions hold throughout $\lambda \in [0, 1]$. This explains the observed lower sensitivity to fixed learning rates. Changing a and b in NEG- t -SNE reintroduces UMAP-like numerical challenges, and the same analysis also shows that NEG- t -SNE and Parametric UMAP are equivalent under the stated formulation.

Overall, the reported comparisons, metric trends, and controlled shape modifications point to the same conclusion: attraction primarily governs whether clusters form and whether runs converge consistently, while repulsion primarily adjusts margins and compactness. This evidence-based framing provides practical tools for optimizing dimensionality reduction algorithms and for interpreting the embeddings they produce. Because dimensionality reduction is closely connected to contrastive learning (Damrich and Hamprecht, 2021; Hu et al., 2023), the same shape-based reasoning may also inform representation learning. Taken together, the analysis makes UMAP-style embeddings more principled, more consistent, and easier to evaluate quantitatively.

Software and Data

All datasets used for the reported comparisons are publicly available. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com. Single-cell transcriptomes data is available at github.com.

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Appendix

The appendix provides the derivations, additional comparisons, and implementation details that support the quantitative framing in the main text. Appendix A derives UMAP’s attraction and repulsion shapes. Appendix B revisits constant learning rates, and Appendix C applies the attraction-repulsion interpretation to large language model embeddings. Appendix D.1 defines the Procrustes distance and matrix used to quantify run-to-run consistency, and Appendix E repeats the random-initialization experiment on additional datasets. Appendix F extends the NEG- t -SNE and UMAP comparison, Appendix G compares additional dimensionality reduction methods, and Appendix H summarizes how the high-dimensional graphs are constructed. Appendix I reports detailed sweeps of λ_a and λ_r (from Fig. 1) for UMAP, NEG- t -SNE, and PaCMAP across datasets, Appendix J tests mixed attraction and repulsion shapes, and Appendix K gives implementation details.

A Proofs and Derivations

A.1 Proof of Proposition 4.1

Proof. The contraction margin follows by subtracting Eq. (7) from Eq. (8) and taking a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

Thus, the pairwise distance contracts exactly when $|1 + 2\lambda f_a| < 1$, i.e., when

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition 4.2

Proof. The expansion margin follows by subtracting Eq. (9) from Eq. (10) and taking a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

The distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

Because repulsion shapes are defined with $f_r > 0$ in the cases studied here, the latter condition suffices. □

A.3 Attraction Shape:

To derive a comparable attraction shape, we use the general low-dimensional affinity $q_{ij} = (\gamma + a\|y_i - y_j\|_1^{2b})^{-1}$. This reduces to UMAP when $\gamma = 1$ and to NEG- t -SNE when $\gamma = 2$. The attractive force is

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

With $\zeta = \|y_i - y_j\|_2$, the scalar coefficient gives the attraction shape:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

The distance dependence of f_a^U is characterized by its derivative:

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

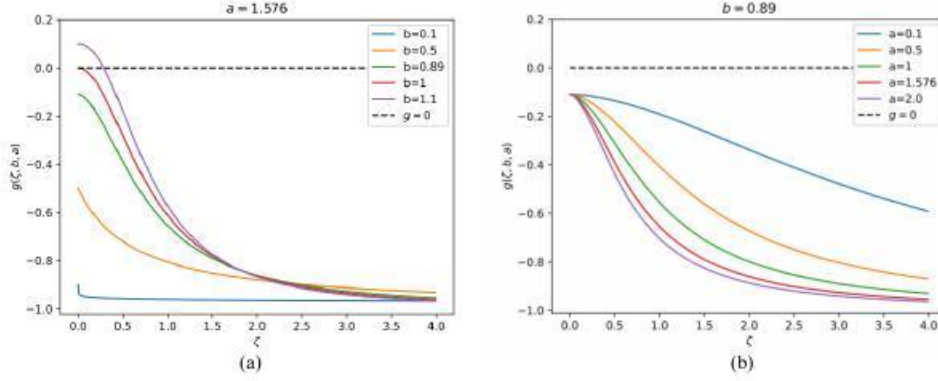


Figure 5: Strict-increase diagnostic for the UMAP attraction shape. Values of $g(\zeta, b, a)$ are shown with (a) a fixed at 1.576 and (b) b fixed at 0.89.

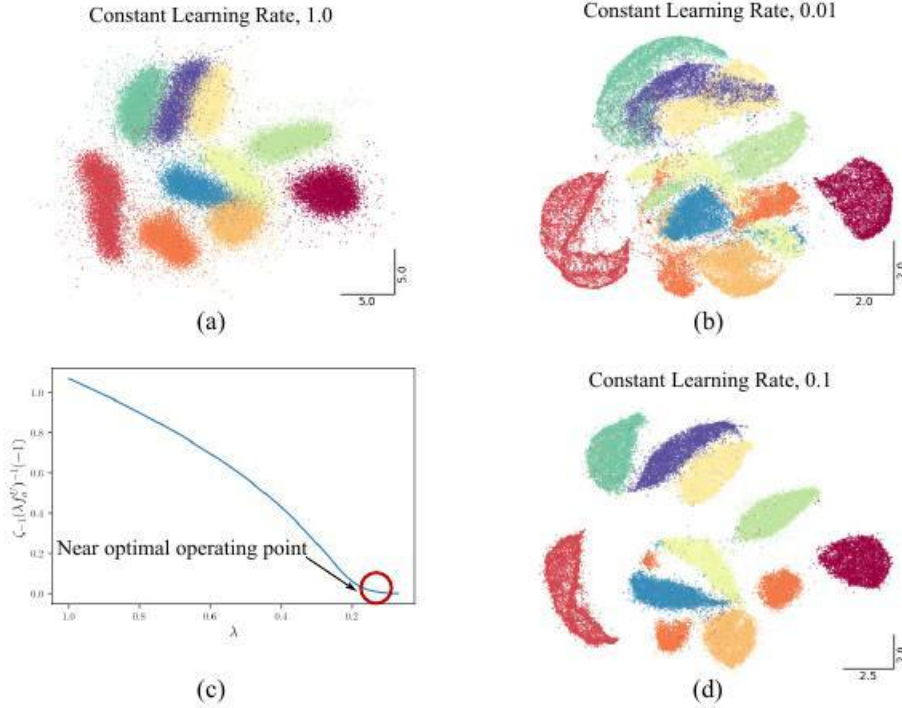


Figure 6: Effect of constant learning rate in embeddings. (a) When the learning rate is too high ($\lambda = 1.0$), the embeddings are diffuse (because of the high value of ζ_{-1}). (b) When the learning rate is too low ($\lambda = 0.01$), clusters do not form (the strengths of attraction and repulsion are too low). (c) ζ_{-1} decreases nonlinearly as the learning rate decreases. The goal of the algorithm is to reduce ζ_{-1} while keeping effective levels of attraction and repulsion. (d) Distinct and compact clusters form at a constant, near-optimal learning rate $\lambda = 0.1$.

This derivative gives the strict-increase condition ($\frac{df_a^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. The inequality holds for $0 < b \leq 1$ by the derivative and asymptotes of g . Figure 5 visualizes this diagnostic across b and a values; once $b > 1$, the monotonicity condition in (23) no longer holds.

A.5 Repulsion Shape:

Using the same general affinity, the repulsive force is

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The scalar coefficient gives the repulsion shape:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

For the parameter regimes considered, $f_r > 0$, matching the expansion condition.

A.6 Loss functions due to modified attraction and repulsion shape:

The attraction-shape modification in Eq. (13) corresponds to the following attractive cost term:

$$- \int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

and the repulsion-shape modification in Eq. (15) corresponds to

$$- \int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the added term can be read as a norm-based regularizer. Expressing the same change at the level of shapes makes the comparison clearer: the attraction modification increases far-sightedness, while the repulsion modification adds a constant repulsive coefficient.

B More on UMAP’s Learning Rate

Sections 5 and 6 explain UMAP’s dependence on learning-rate annealing through a measurable contraction margin (Fig 6). In Section 5, the minimum contraction distance ζ_{-1} quantifies how much of the short-distance range produces expansion rather than contraction; reducing this value through annealing yields more compact clusters. Section 6 then compares UMAP with $a = 1.0$ and $b = 1.0$ against NEG- t -SNE: NEG- t -SNE tolerates a constant learning rate of 1.0 better because its attraction shape lies in $[-1, 0]$, whereas UMAP’s lies in $[-2, 0]$. The same margin argument predicts that UMAP should tolerate a constant learning rate of 0.5, which confines its effective attraction to $[-1, 0]$ (Figs. 11(f,h)).

Across a wide range of a and b , annealing still gives stronger embeddings because it eventually drives ζ_{-1} toward zero as the learning rate approaches zero. Without that reduction, the embedding remains diffuse (Fig 6(a)). Conversely, if the learning rate starts too low, both attraction and repulsion are too weak to form clear clusters (Fig 6(b)). The ζ_{-1} versus λ curve in Fig 6(c) identifies $\lambda = 0.1$ as a favorable constant-rate compromise: ζ_{-1} is low while attraction and repulsion remain substantial. With this constant learning rate, the embedding has compact clusters and clear boundaries (Fig 6(d)).

C Large Language Model Embedding

The same attraction-repulsion framing also improves a large text-embedding visualization. We randomly selected 300,000 PubMedBERT (Gu et al. 2021) feature vectors with known labels from the PubMed dataset (González-Márquez et al. 2024). Default UMAP produces a crowded map in which label colors overlap substantially (Fig. 7(a)), and without labels the cluster structure is difficult to interpret. Reducing b to 0.6, following the analysis in Section 5.2, increases repulsion while also strengthening attraction-driven cluster formation. The resulting map is more explorable even without explicit labels (Fig. 7(b)). The practical gain is also visible in the reported metric: on a 10,000-point hold-out set with $k = 10$, k-NN accuracy increases from 49.2% under default UMAP to 55.48% after reducing b to 0.6.

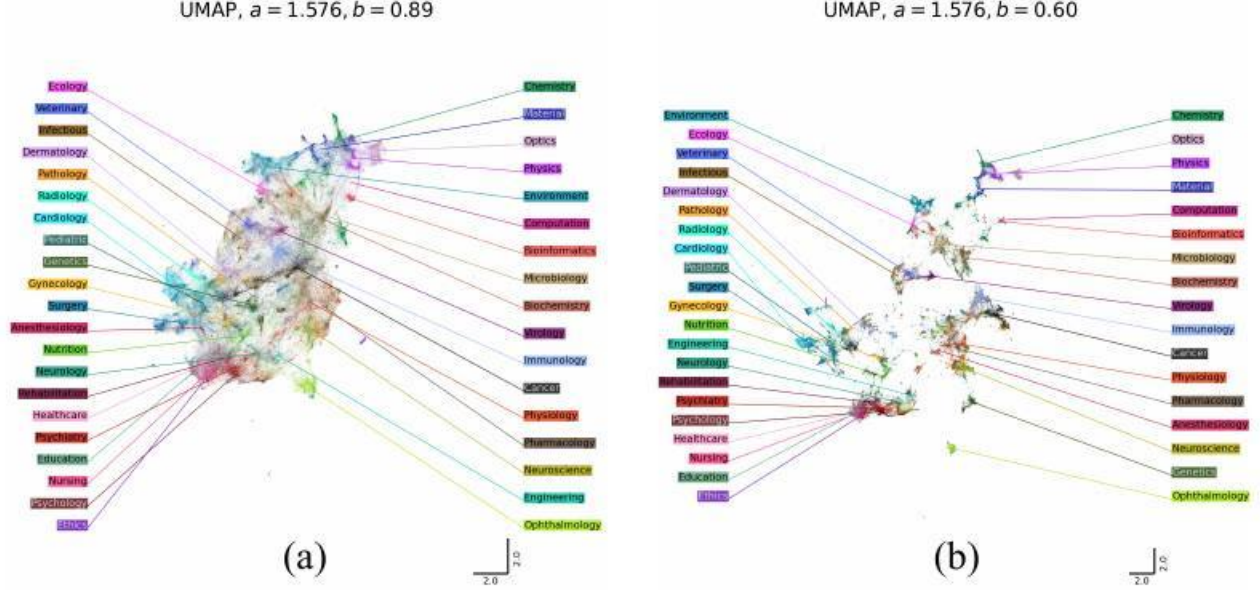


Figure 7: PubMedBERT embedding comparison for 300K PubMed abstracts. (a) Default UMAP ($a = 1.576, b = 0.89$) produces a crowded map with overlapping labels. (b) Increasing attraction and repulsion through $b = 0.60$ yields clearer cluster formation and more separated labels. Labels are shown in both panels to make the comparison readable.

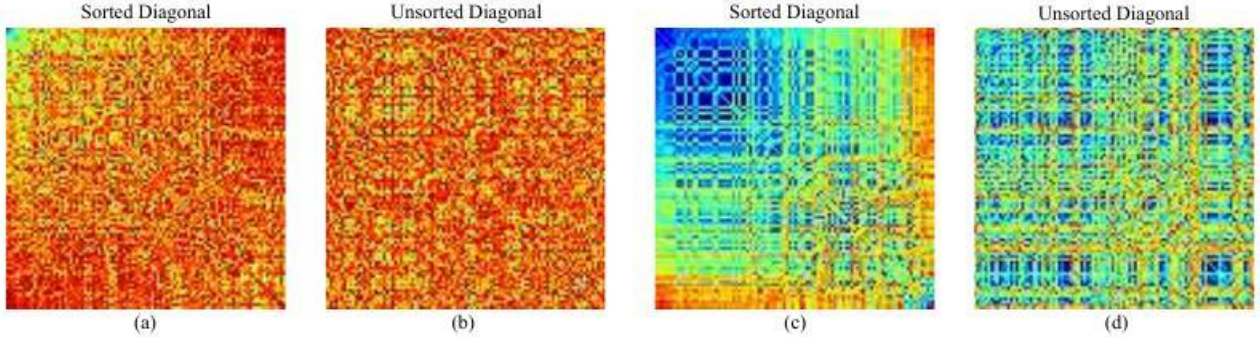


Figure 8: Procrustes-matrix visualization with and without sorted reference distances. (a) Matrix from Fig. 2f for default UMAP, with the diagonal sorted by p_d to the PCA-initialized reference. (b) The same values as (a) with an unsorted/random diagonal. (c) Matrix from Fig. 2g for the modified attraction shape, again sorted by reference p_d . (d) The same values as (c) without sorting. Sorting does not change the distances but makes run-to-run self-similarity easier to see.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower [1975]) provides the quantitative consistency metric used in the random-initialization experiments. It measures similarity between two point clouds $\{x\}$ and $\{y\}$ after allowing translation, scaling, and rotation. Operationally, $\{x\}$ is held fixed while $\{y\}$ is transformed to maximize alignment. If $\{y'\}$ is the aligned version of $\{y\}$, then the Procrustes distance is

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

This linear distance has been useful in multiple embedding-analysis settings (McInnes et al. [2018] Islam and Fleischer [2022] Kotlov et al. [2024]).

For our experiments, Procrustes distance measures how consistently an embedding method reaches the same configuration from random initialization. Let $\{x\}_p$ denote the PCA-initialized reference embedding, and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be the N embeddings produced from random starts. We summarize pairwise run similarity by the mean and standard deviation of the strictly lower-triangular entries of the matrix P (reported as mean $\pm$ std in Figs. 2, 9 and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

To make the comparison visually ordered, the indexes of X_r can be sorted so that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$. The diagonal values of the Procrustes matrix are then

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal does not change any Procrustes distances, but it makes the empirical pattern easier to interpret. In Fig. 8(a), the sorted default-UMAP matrix groups similar runs in the upper-left region. With the modified attraction shape, the larger blue region in Fig. 8(c) shows that more random-initialized outputs are mutually similar, consistent with a stronger metastable configuration. When the same matrices are unsorted, this structure is visually obscured even though the underlying distances are unchanged (Fig. 8(b,d)).

D.2 Trustworthiness

Trustworthiness (Venna and Kaski, 2001) is the neighborhood-preservation metric used in the learning-rate sweeps. It quantifies how well local neighborhoods survive dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n-3k-1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. We use this metric to compare embeddings on the same local-neighborhood criterion. In practice, k is often set to 5, assessing preservation of each point’s five nearest neighbors. For computational efficiency, each reported trustworthiness value uses 10,000 randomly sampled indices, and comparisons reuse the same indices across embeddings.

D.3 Silhouette Score

Silhouette score (Rousseeuw, 1987) is the label-separation metric used alongside trustworthiness. Although originally designed for clustering algorithms, here it measures how clearly ground-truth labels separate in the embedding. For a point y_i in point cloud $\{y\}$, let a_i be the mean distance from y_i to points with the same label, and let b_i be the smallest mean distance from y_i to points with any other label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies in $[-1, 1]$: values near 1 indicate strong placement within the correct label cluster, values near 0 indicate boundary points, and values near -1 indicate poor label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We compute silhouette score on the full embedding, without random sampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

Section 5.1 reports the random-initialization experiment on MNIST; here we repeat the same 100-run comparison on Fashion-MNIST (FMNIST) (Xiao et al., 2017) and single-cell transcriptomes (Macosko et al., 2015) (Fig. 9 and 10 respectively). The metric pattern is consistent across datasets: modified and composite attraction shapes that increase long-distance attraction reduce Procrustes distances and improve reconstruction consistency.

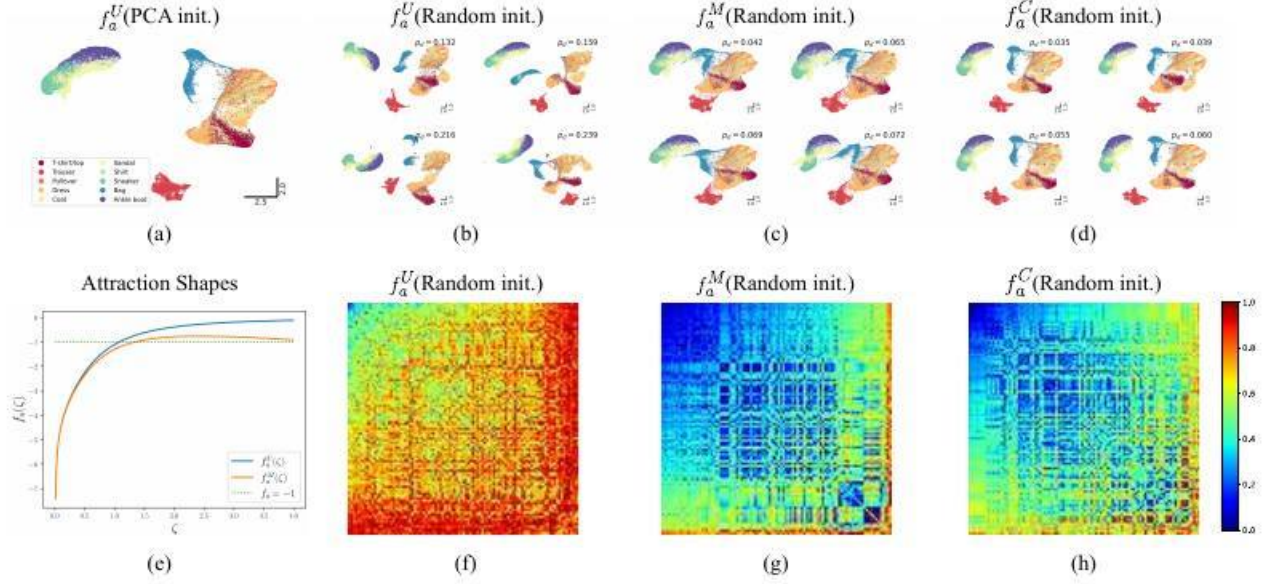


Figure 9: FMNIST random-initialization consistency across attraction shapes. (a) PCA-initialized reference mapping. (b-d) Four random-initialized mappings with the lowest Procrustes distance (p_d) to (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18), showing lower mean distances for the modified and composite settings.

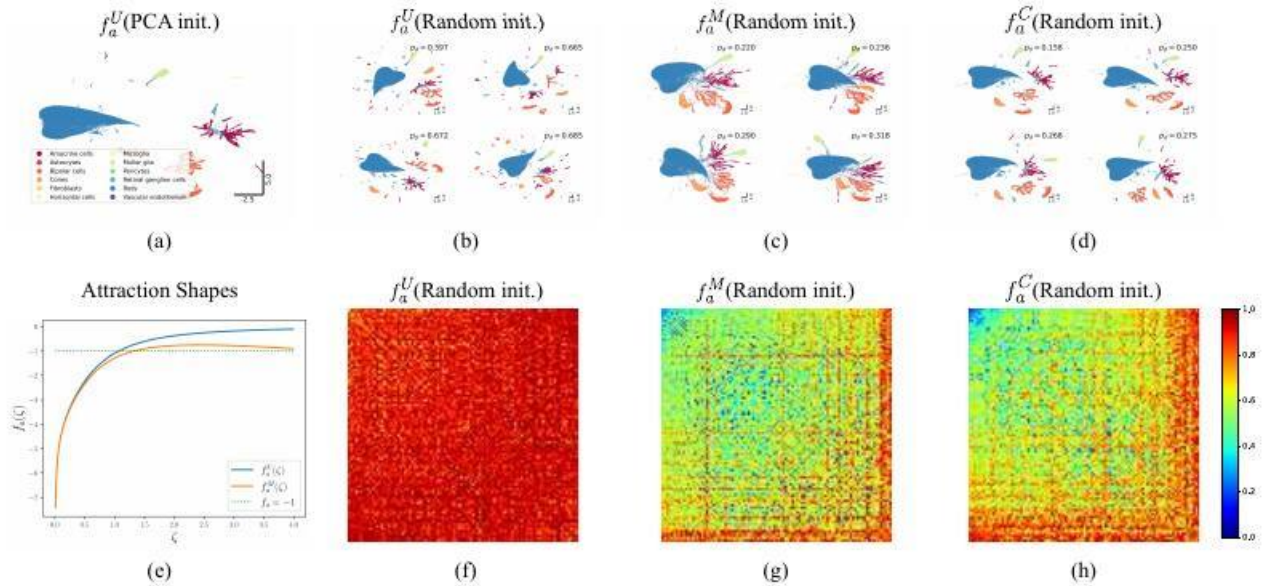


Figure 10: Single-cell transcriptomes random-initialization consistency across attraction shapes. (a) PCA-initialized reference mapping. (b-d) Four random-initialized mappings with the lowest Procrustes distance (p_d) to (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15), again indicating improved consistency after strengthening attraction at large distances.

F More on NEG- t -SNE

F.1 Comparison to UMAP

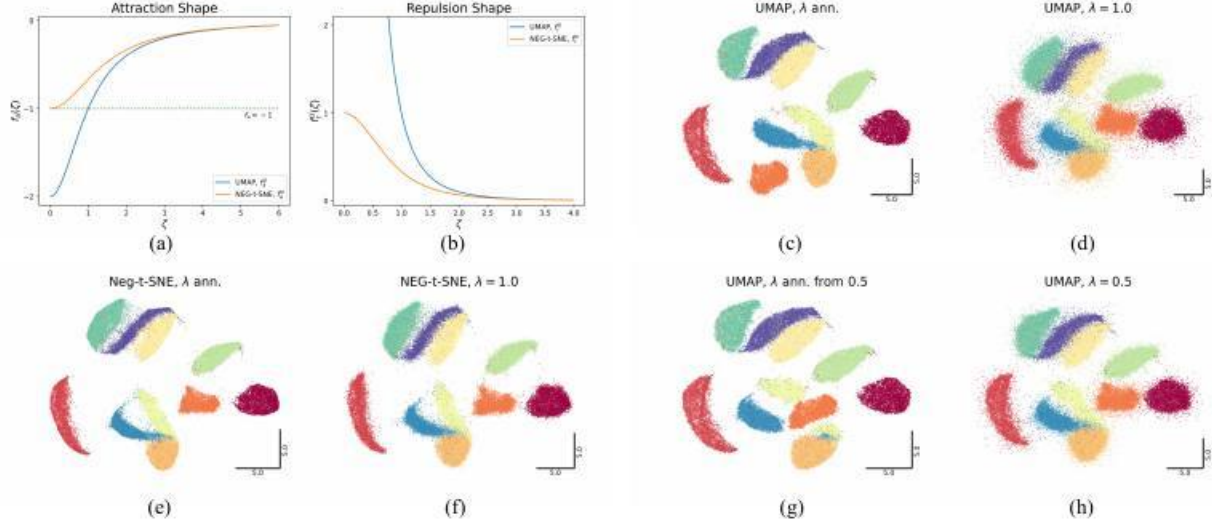


Figure 11: Learning-rate sensitivity comparison on MNIST for UMAP and NEG- t -SNE. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1, b = 1$) and NEG- t -SNE. (c,d) UMAP becomes fuzzy under constant $\lambda = 1$ because $f_a^U < -1$ at small ζ , placing close pairs outside the contraction margin. (e,f) NEG- t -SNE keeps $f_a^N \in [-1, 0]$, so its clusters remain cleaner without annealing. (g,h) Setting $\lambda = 0.5$ confines UMAP’s effective attraction to $[-1, 0]$ and reduces sensitivity.

Figure 11 compares UMAP and NEG- t -SNE shapes with the corresponding MNIST embeddings. With annealing, the two methods produce similar outputs (Figs. 11 c,e). Under the stricter condition of constant $\lambda = 1$, UMAP becomes visibly fuzzy, whereas NEG- t -SNE retains cleaner cluster boundaries (Fig. 11 d,f). Section 6 predicts this pattern from the attraction margin: keeping f_a^U within $[-1, 0]$ should reduce fuzziness at fixed λ . The metric sweeps in Fig. 11 h,i show the same trend, with better UMAP embedding and clustering when attraction learning rate $\lambda_a \lesssim 0.5$. Setting $\lambda = 0.5$ satisfies Proposition 4.1 for all ζ , and the resulting embeddings (Figs. 11 g,h) show sharper boundaries than constant $\lambda = 1$ while resembling NEG- t -SNE clusters. A few points remain outside clusters because UMAP’s repulsion shape differs from the bounded NEG- t -SNE repulsion.

Introducing parameters a and b into NEG- t -SNE gives the Parametric UMAP formulation in Proposition 6.1. The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, with attraction and repulsion shapes $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$ and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes become unbounded as $\zeta \rightarrow 0$, so varying a and b reintroduces UMAP-like numerical challenges. The main favorable difference is that the attraction shape reaches a lower minimum contraction distance ζ_{-1} than UMAP, which can support cluster formation. At the same time, the attraction approaches zero more quickly at large distances, increasing near-sightedness and potentially reducing long-range attraction.

F.2 Comparison to Parametric UMAP

Parametric UMAP was initially trained with the original UMAP objective (Sainburg et al. 2021); later work adopted a numerically stable, log-sigmoid-based modified cross-entropy loss (Shi et al. 2023). Under this loss, the comparison is exact: Parametric UMAP and Neg- t -SNE (Damrich et al. 2023) are equivalent as stated in Proposition 6.1. The algebra below preserves the parameters that make the equivalence explicit.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Together, these two terms are exactly NEG-t-SNE with explicit parameters a and b ; standard Neg- t -SNE sets both parameters to 1. $\square$

G Alternate Dimensionality Reduction Algorithms

The alternative algorithms considered here share the same low-dimensional kernel as UMAP with $a = 1$ and $b = 1$, which makes their force shapes directly comparable:

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We first discuss TriMap (Amid and Warmuth, 2019). Although TriMap uses triplets rather than pairwise interactions, it provides a useful comparison case because its attraction and repulsion terms are coupled. We then examine Pairwise Controlled Manifold Approximation (PaCMap) (Wang et al., 2021) and its extension Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al., 2025), both of which adapt TriMap-style ideas to pairwise interactions. The section then compares t -SNE (Van der Maaten and Hinton, 2008), briefly notes SNE (Hinton and Roweis, 2002) with its alternate kernel, and concludes with multidimensional scaling (Borg and Groenen, 2007).

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes the low-dimensional embedding with a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} is the triplet weight, y_j is a high-dimensional k -nearest neighbor of y_i , and y_k is a far-away point. Minimizing this loss should decrease the distance between y_i and y_j while increasing the distance between y_i and y_k . The update equations make the coupled comparison explicit:

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

where $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the far-point distance, and f_a^T and f_r^T are TriMap's attraction and repulsion shapes. Unlike UMAP, TriMap does not separate the attractive and repulsive components; both shapes depend on two distances, so the comparison is two-dimensional. The attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

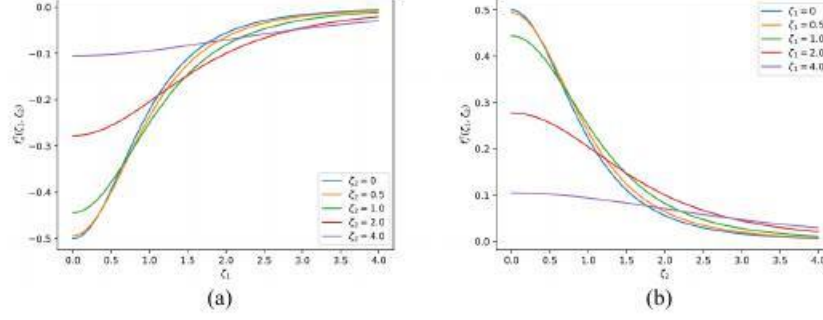


Figure 12: TriMap shape comparison. (a) Attraction shapes across values of ζ_2 . (b) Repulsion shapes across values of ζ_1 , showing how each force depends on the coupled triplet distances.

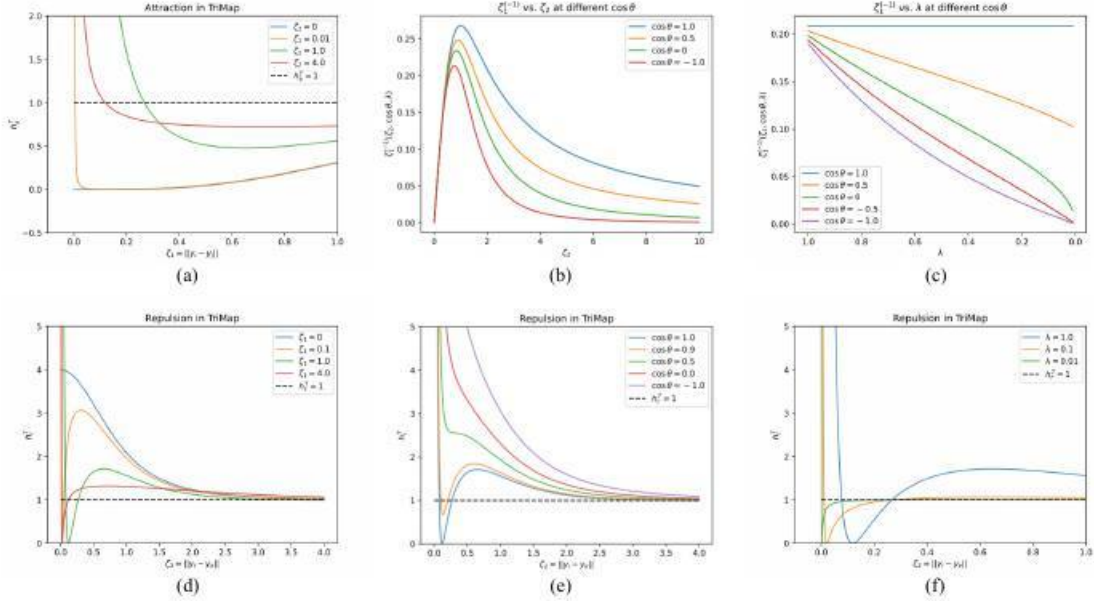


Figure 13: TriMap contraction and expansion margins. (a) h_a^T versus ζ_1 for different ζ_2 values; values below the dotted line indicate attraction, so TriMap, like UMAP, can show both contraction and expansion for nearest-neighbor pairs (y_i, y_j) . (b,c) The minimum contraction distance ($\zeta_1^{(-1)}$) varies with (b) ζ_2 and (c) λ , with $\cos \theta$ regulating the range. (d-f) Repulsion behavior as (d) ζ_1 , (e) $\cos \theta$, and (f) λ vary; values above (below) the dotted line indicate repulsion (attraction). Unlike UMAP's purely repulsive negative-edge force, TriMap repulsion can also contract. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

TriMap's shapes in Fig. 12 follow trends similar to UMAP, but the attraction-side minimum is -0.5 , so attraction alone does not flip point positions. Because Eqs. 45-47 couple attraction and repulsion, Propositions 4.1 and 4.2 do not apply directly. The contraction condition therefore has to include the coupled geometry:

Proposition G.1. Update equations 45-47 provide a contraction if

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. 45 and 46: $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. 51, we obtain the desired inequality. $\square$

Inequality 50 depends jointly on ζ_1 , ζ_2 , $\cos \theta$, and λ . Thus, the nearest-neighbor distance $\zeta_1 = \|y_i - y_j\|_2$ cannot be judged independently of the far-point distance and angle. Figure 13(a) fixes $\cos \theta = 1$ and $\lambda = 1$ and varies ζ_2 ; values below the dotted line indicate contraction of ζ_1 , and values above indicate expansion. The intersection with the dotted line defines a minimum contraction distance $\zeta_1^{(-1)}$, analogous to UMAP's ζ_{-1} :

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

For most settings, $\zeta_1^{(-1)}$ is finite and positive (Fig. 13(b,c)), so TriMap can share UMAP's need for a mechanism that controls the contraction margin. In the TriMap implementation, this role is served by delta-bar-delta (Jacobs 1988) together with appropriate initialization.

Proposition G.2. Update equations 45-47 provide expansion if

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. 45 and 47: $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. 55 we obtain inequality 54. $\square$

The repulsion condition in Inequality 54 has the same dependence on ζ_1 , ζ_2 , $\cos \theta$, and λ . Here the target quantity is expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) vary the other quantities and show the margin: values above the dotted line expand distance, and values below contract it. The important contrast with UMAP is that TriMap's repulsive component can sometimes act attractively. Because this occurs at small distances, suitable initialization and triplet selection can reduce the risk.

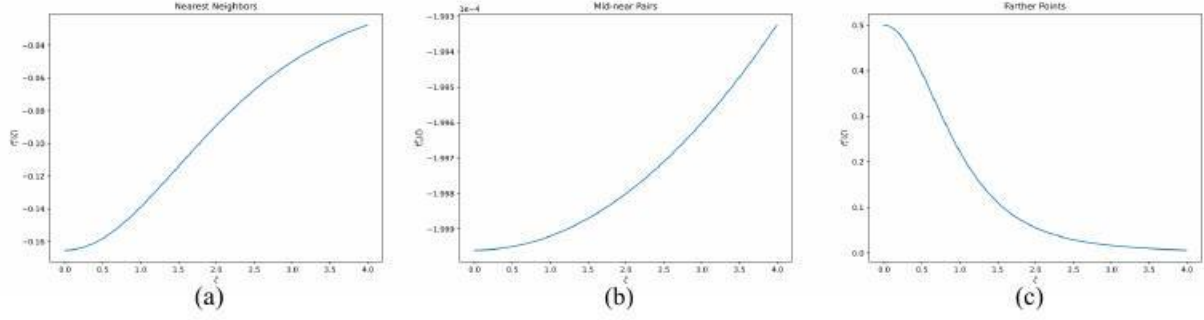


Figure 14: PaCMAP shape comparison. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near pairs, with mid-near values on the order of 10^{-4} . (c) Repulsion shape for farther pairs. All panels use $\lambda = 1$.

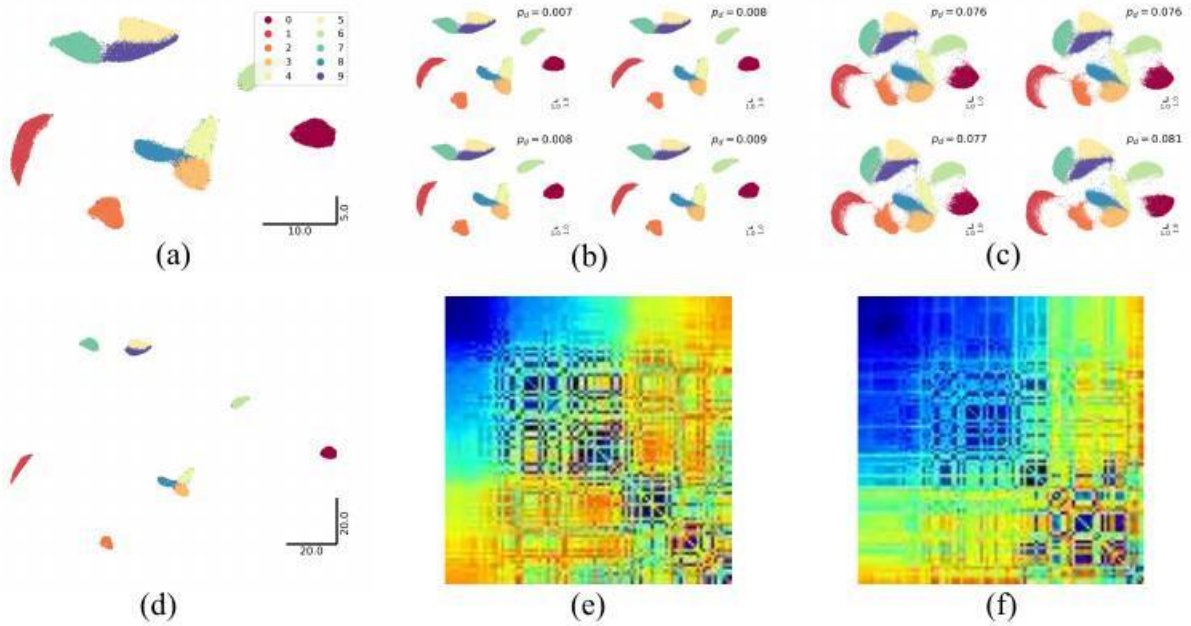


Figure 15: PaCMAP behavior for different conditions. (a) PaCMAP of MNIST data with PCA initialization. (b,c) Four samples that best match with (a) by (b) original PaCMAP and (c) PaCMAP with modified attraction shape, $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$, when randomly initialized. (d) PaCMAP of MNIST with a modified repulsion shape $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrix for (e) original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with modified attraction shape (0.43 ± 0.22). Similar to UMAP, increased attraction at farther distances show improved consistency, while increased repulsion shows smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes low-dimensional embedding at different scales. The loss function is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} are weights of the nearest neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (details in Appendix H). The first two terms provide attraction, whereas the last term provides repulsion. A closer look at the loss function reveals that the function is a modified form of TriMap’s triplet loss. For the attractive terms, it replaces TriMap’s affinity for distant points with constant terms (1/10 for nearest neighbors, 1/10000 for mid-near pairs, and 1 for farther points). This loss function is thus separable into three terms and decouples

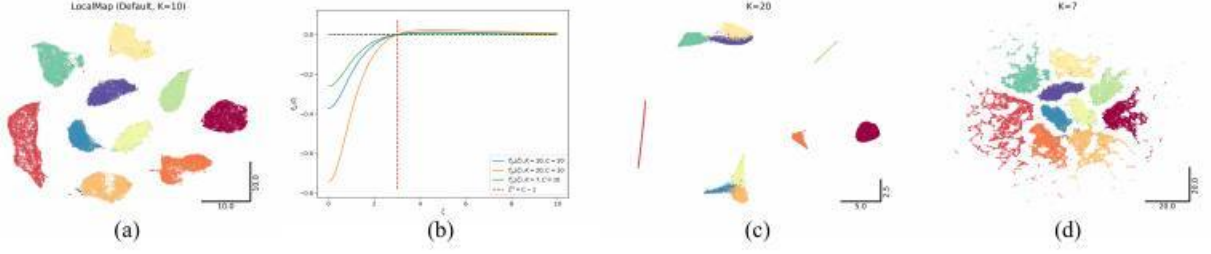


Figure 16: LocalMAP behavior on MNIST. (a) Default embedding. (b) Nearest-neighbor attraction-repulsion shape; values below (above) the dotted line indicate attraction (repulsion), and the transition occurs when ζ crosses $\sqrt{C-1}$. Following LocalMAP’s implementation, $C = 10$. (c) With larger K ($= 20$), repulsion dominates and clusters become compact. (d) With smaller K ($= 7$), attraction dominates and existing clusters break up.

the update equations. The update equations of the nearest neighbor term are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

of the mid-near pairs are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and of the farthest pairs are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, $\zeta_3 = \|y_i - y_l\|_2$ are distances, f_a^P and f_m^P are attraction shapes for nearest neighbors and mid-near pairs, respectively, and f_r^P is the repulsion shape for the farthest pairs. Correspondingly, the functional forms of the shapes are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P satisfy Proposition 4.1, and f_r satisfies Proposition 4.2 (Fig. 14). PaCMAP’s attraction is smaller than UMAP’s, but remains effective across a wide learning-rate range when modulated by Adam (Kingma and Ba, 2015); with $w_{NB} = 3$, the maximum attraction stays below 0.5, preventing flips during attractive updates. PaCMAP typically initializes points in a small low-dimensional sphere (on the order of 10^{-3} in (Wang et al., 2021)) and then relies on repulsion to separate clusters. Empirically, it largely recovers UMAP-like clustering, especially on MNIST, while improving ordering. Its random-initialization consistency is better than UMAP’s (Fig. 15(b,e)) and improves further with modified attraction (Fig. 15(c,f)). Adding a small positive value to the repulsion shape increases compactness by increasing inter-cluster distance.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al., 2025) extends PaCMAP and provides another useful comparison because it separates all 10 MNIST clusters (Fig. 16(a)). Its behavior resembles the compactness and split-cluster patterns in Figs. 3(g,h,j,k) and Figs. 4(a,b), so it helps quantify how attraction and repulsion jointly affect cluster compactness and connectedness.

LocalMAP first performs PaCMAP and then adds an optimization stage that can decouple clusters through a nearest-neighbor attraction-repulsion term. It minimizes

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term combines attraction and repulsion on the same nearest-neighbor pair: one distance regime contracts, and the other expands. The second term matches PaCMAP’s far-pair term, except that LocalMAP resamples far pairs every few iterations. We therefore analyze the nearest-neighbor term, whose update equations are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_1^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update provides contraction as long as $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ (from Proposition 4.1). When $\zeta^2 > C-1$, $f_{ar} > 0$, and by Proposition 4.2 the update equations provide expansion. The parameters K and C determine which side of this attraction-repulsion balance dominates. In the LocalMAP implementation, both are set to 10 (Fig. 16(b)), producing stronger attraction than PaCMAP (Fig. 14(a)). For $\zeta > 3$, nearest-neighbor pairs become repulsive, so bridge pairs between clusters separate. The scaling parameter K controls the strength of this separation.

Using Proposition 4.1 the condition $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the largest K that avoids pair flipping under the maximum attraction of f_{ar}^L . Avoiding flips preserves the ordering inherited from PaCMAP and supports cluster separation. This simplifies (69) to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which gives $K \lesssim 13.44$ for $C = 10$ and $\lambda = 1$. For larger $K \gtrsim 13.44$, repulsion dominates and clusters become compact, but the LocalMAP objective fails because the bridge between clusters persists (Fig. 16(c) for $K = 20$). For smaller K , attraction dominates because repulsion is too weak, and existing clusters break apart (Fig. 16(d) for $K = 7$), matching the attraction-dominated behavior in Fig. 4(b).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton 2008) optimizes pairwise similarities, giving another baseline for comparing attraction and repulsion shapes. Its loss is

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the pair weight. The original implementation considers all pairs rather than only nearest neighbors, so the comparison includes global normalization. The loss decomposes into attractive and repulsive terms as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term gives attraction and the second gives repulsion. The attractive term matches UMAP with $a = 1$ and $b = 1$ and is straightforward to evaluate, whereas the repulsive term couples every pair and is therefore costly. Applying the same shape decomposition used for UMAP gives the t -SNE updates. Because original t -SNE does not rely on a nearest-neighbor graph, the pair weight $w_{i,j}$ remains central in every update. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the repulsion shape of t -SNE and $Z = \sum_{k \neq l} q_{k,l}$. The functional forms of these shapes are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1+\zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1+\zeta^2)^2}. \quad (77)$$

From the attractive updates in Eqs. (72)-(73), f_a^{t-SNE} satisfies Proposition 4.1 when $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$, yielding a minimum contraction distance ζ_{-1} . The repulsive update is coupled across all pairs, so a one-dimensional repulsion-shape condition is insufficient. Following the TriMap analysis, the expansion condition becomes

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k,k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l,l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Because the t -SNE repulsion condition in Eq. (78) has the same coupled form as TriMap's, the qualitative behavior is similar: a nominally repulsive update can either expand or contract a pair depending on the surrounding configuration.

Because t -SNE scales attraction by $w_{i,j}$ and repulsion by $\frac{w_{i,j}}{Z}$, its effective forces are typically weaker than UMAP's. This explains why implementations often use much larger learning rates, such as $\approx 10^3$ in Open- t -SNE (Poličar et al. 2024). Most implementations also use early exaggeration, multiplying attractive forces by a constant during the first iterations. Quantitatively, this increases far-sightedness: pairs that should be close but start far apart receive stronger attraction. It can also help separate pairs that are already near $\zeta_{i,j} \approx \zeta_{-1}$ but should not remain together. Early exaggeration is therefore a major contributor to consistent t -SNE embeddings, particularly under random initialization. When applied throughout optimization, the same mechanism makes t -SNE embeddings resemble UMAP more closely (Böhm et al. 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis, 2002) is one of the earliest methods in this family. It follows the same loss form as t -SNE in Eq. (71) but uses the Gaussian low-dimensional affinity $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. The resulting attraction shape is

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

that follows Proposition G.3 after replacing the t -SNE symbols with their SNE counterparts. This attraction shape relies heavily on the learning rate λ and weights $w_{i,j}$ to stay in the contraction regime, which helps explain why clusters can overlap even for small samples—the crowding problem (Van der Maaten and Hinton, 2008). t -SNE and later methods improve this behavior by replacing the Gaussian kernel with the heavy-tailed kernel in Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen, 2007) usually relies on stress majorization because direct gradient methods often fail to produce good mappings. Some gradient-based formulations have nevertheless been studied (Kruskal, 1964; Zheng et al., 2018), and Zheng et al. (2018) gives a successful graph-drawing variant. To compare MDS within the same shape framework, we start from the cost function, absorbing the weight $w_{i,j}$ into the learning rate for this discussion:

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

This loss has a single term for each pair rather than separate attraction and repulsion terms; the same term can attract or repel depending on the current distance. Because the loss is separable over pairs, one pair can be analyzed directly. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

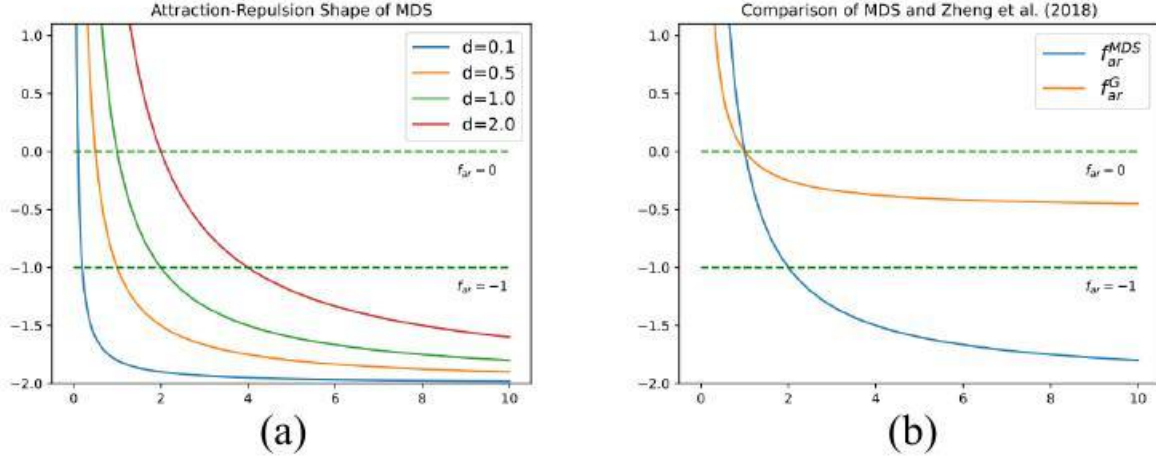


Figure 17: MDS attraction-repulsion characteristics. (a) Shape f_{ar} for standard MDS. (b) Comparison between the MDS shape and the force-directed modification of Zheng et al. (2018) at $d = 1.0$, highlighting the different contraction margins.

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2\frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive when $\zeta > d$ and is bounded within $[-2, 0]$ in that region (Fig. 17(a)). The bound is not automatically favorable: only values in $[-1, 0]$ contract, while values outside that interval expand, and values below -2 flip point positions before expansion. To keep the effective attraction within a safer range, Zheng et al. (2018) uses a force-directed modification:

$$f_{ar}^G(\zeta) = -\frac{1}{2}\frac{\zeta - d}{\zeta}, \quad (85)$$

This modification works for effective learning rates $\lambda \leq 1$ because f_{ar}^G is bounded in $[-0.5, 0]$. For $\zeta < d$, standard $f_{ar}^{MDS} > 0$ is repulsive. Overall, the MDS attraction-repulsion interaction is most effective when initialization is already close to the desired output. Continued pairwise optimization oscillates around $\zeta = d$, where $f_{ar}^G = 0$, giving a target-distance analogue of UMAP's ζ_{-1} . Unlike UMAP, no learning-rate choice reduces this target distance to zero, because preserving it is the MDS objective. This helps explain why clusters are often fuzzier than in UMAP or PaCMAP (for illustrations, see Lambert et al. (2022), de Bodt et al. (2025), and Kury et al. (2025)). Lambert et al. (2022) further converts the MDS loss to a quartet stress with relative distances, enabling global regularization and lower computation than majorization.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) Hinton and Roweis (2002) provides the graph-construction template for many modern dimensionality-reduction algorithms. For dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$, it defines high-dimensional pair weights by

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is selected so that the neighborhood scale matches the user-defined *perplexity* P :

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP instead builds its high-dimensional comparison graph from k-nearest neighbors (k-NN):

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k nearest neighbors of x_i , and σ_i is chosen so that $\sum_j p_{i|j} = \log_2(k)$. The directed scores are then symmetrized by a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses the k-NN graph directly, assigning affinities equal to 1, and combines it with a self-tuning distance measure (Zelnik-Manor and Perona, 2004):

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance from x_i to its fourth through sixth Euclidean nearest neighbors. Although this definition differs from the σ_i terms in Eq. (87) and (90), all three quantities set a local scale for high-dimensional comparisons.

These graph-construction choices define the high-dimensional evidence supplied to the optimizer. Once optimization begins, our analysis shows that the attraction and repulsion shapes are the primary drivers of the resulting low-dimensional geometry.

I Detailed Results for Varying λ_a and λ_b

This section expands the learning-rate evidence summarized in Fig. 11(h,i). Each comparison replaces UMAP’s attraction and repulsion shapes with the corresponding shapes from another method, then sweeps attraction learning rate λ_a and repulsion learning rate λ_r . Figure 18 reproduces the MNIST metric summary from the main text. Figures 19-22 show the individual MNIST embeddings for each λ_a, λ_r setting, together with the initialization, an annealed- λ reference, and the ablations where either λ_a or λ_r is set to 0. For PaCMAP, which includes mid-near pairs, we also show a reference with mid-near pairs included. Figures 23-27 and Figures 28-32 report the same comparison structure for FMNIST and single-cell transcriptomes, respectively.

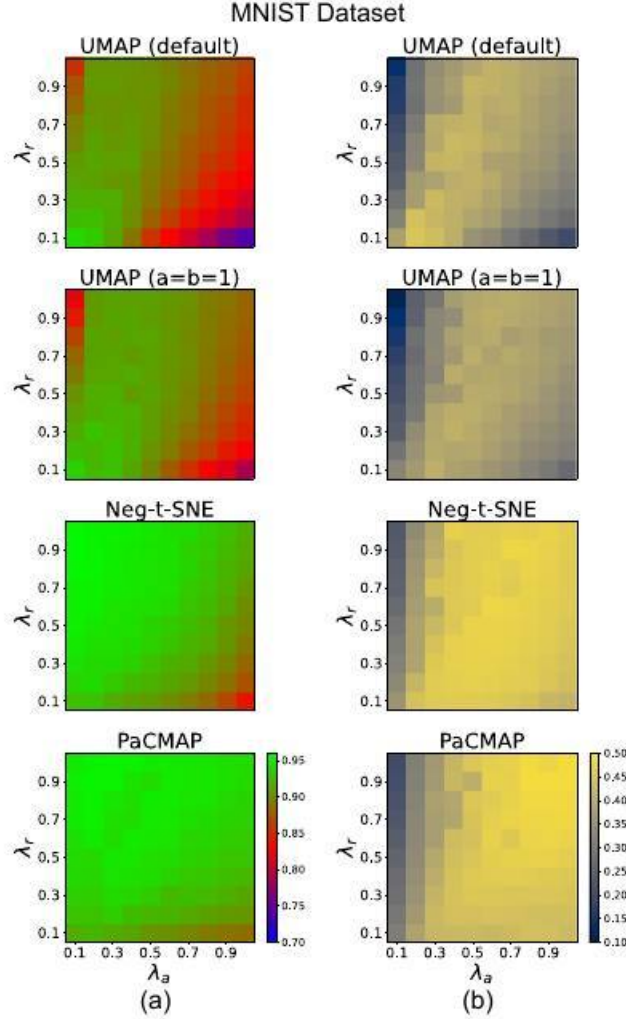


Figure 18: MNIST metric comparison across methods. (a) Trustworthiness measures neighborhood preservation. (b) Silhouette score measures label separation.

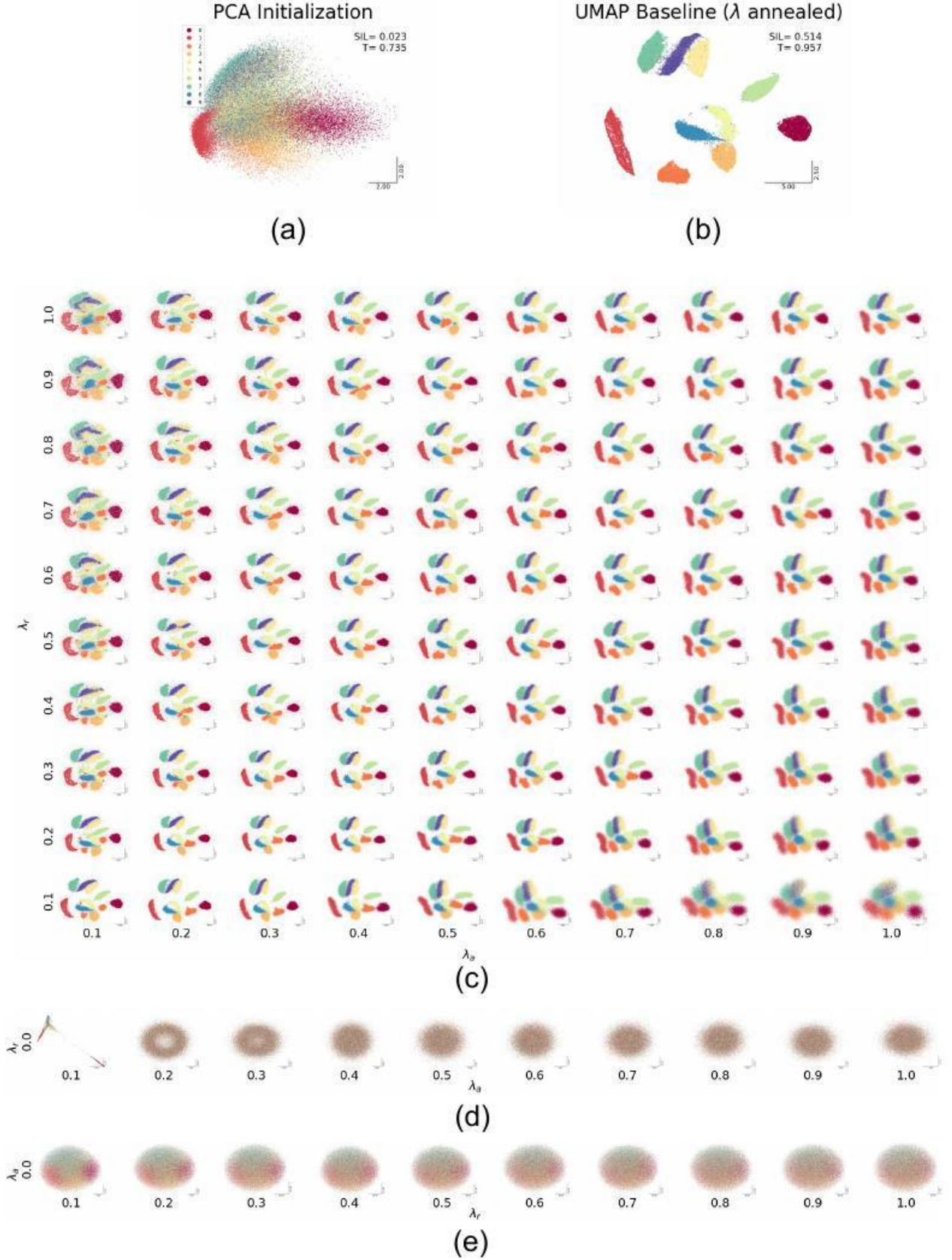


Figure 19: MNIST sweep of λ_a and λ_r using UMAP attraction and repulsion shapes. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) Setting $\lambda_r = 0$ shows that attraction alone does not form clusters. (e) Setting $\lambda_a = 0$ shows that repulsion alone also does not form clusters.

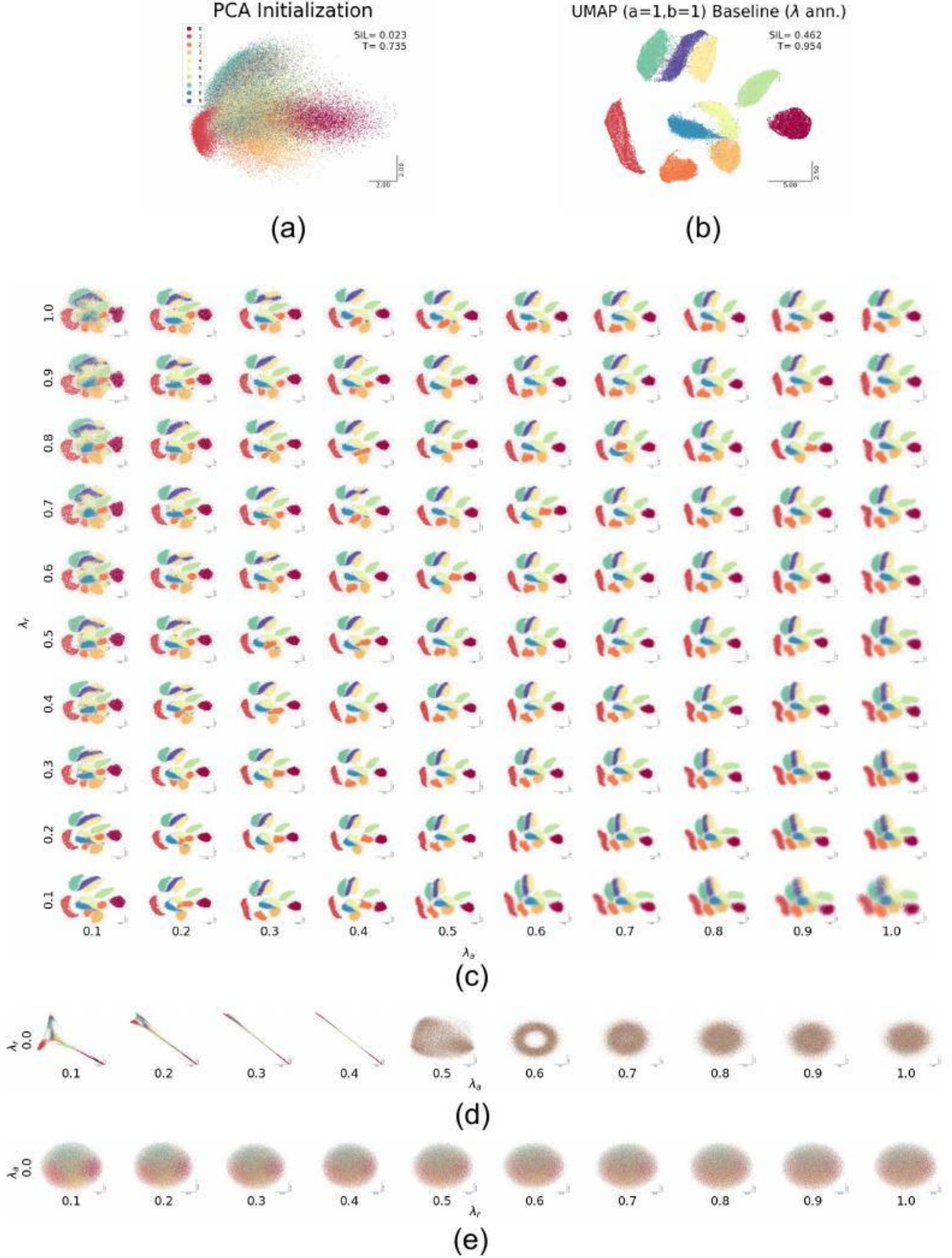


Figure 20: MNIST sweep of λ_a and λ_r using UMAP shapes with $a = 1$ and $b = 1$. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

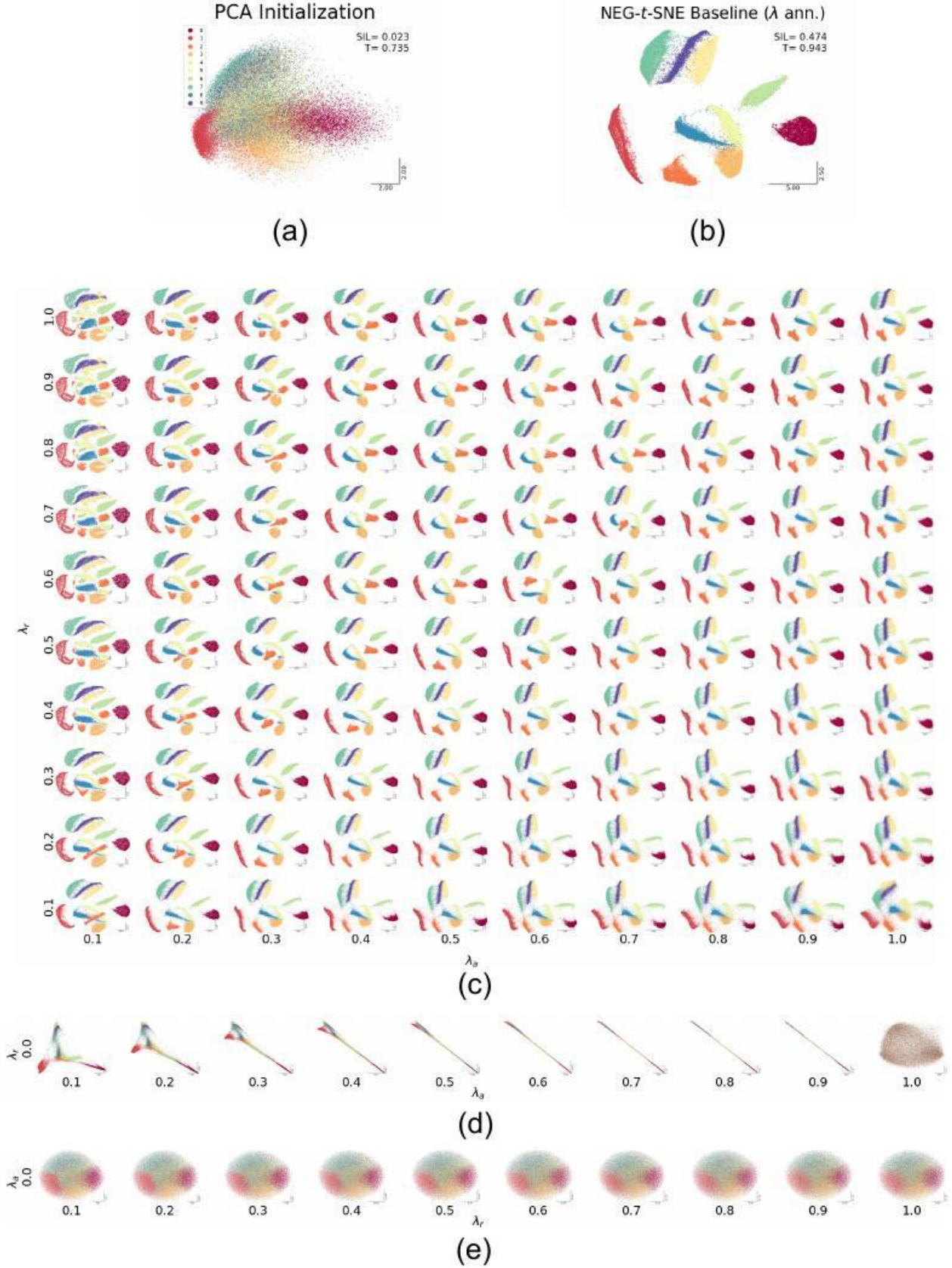


Figure 21: MNIST sweep of λ_a and λ_r using NEG-t-SNE attraction and repulsion shapes. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

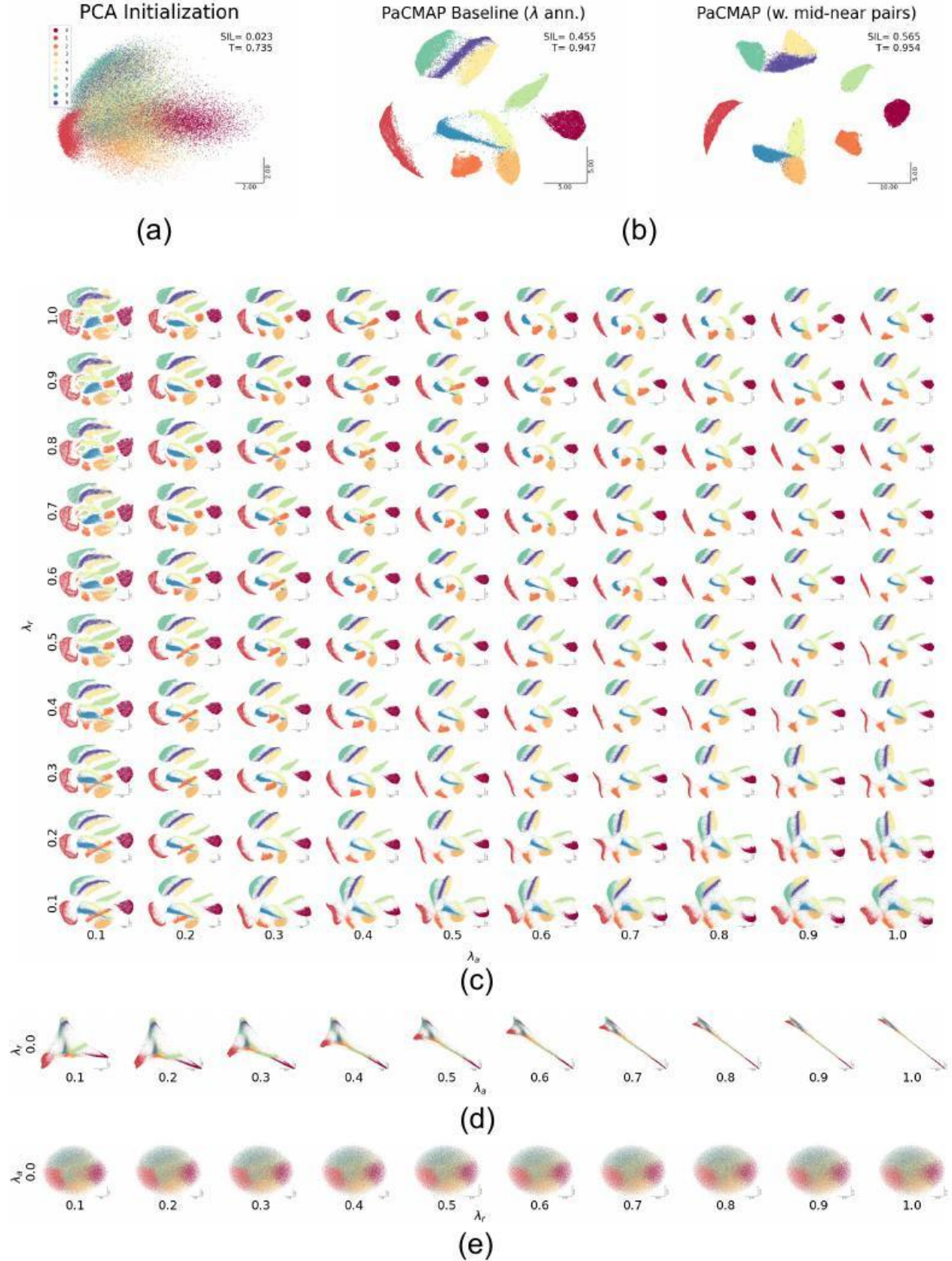


Figure 22: MNIST sweep of λ_a and λ_r using PaCMAP attraction and repulsion shapes. (a) Shared initialization. (b) Left: annealed- λ baseline starting from 1 with mid-near points excluded to isolate attraction-repulsion interaction; right: reference with mid-near points included. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

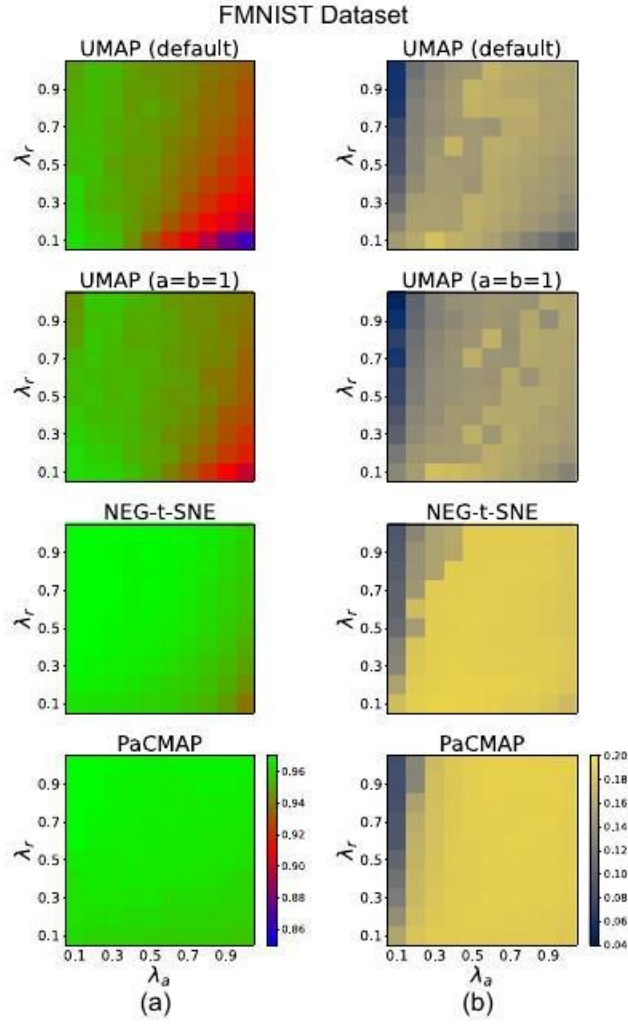


Figure 23: FMNIST metric comparison across methods. (a) Trustworthiness measures neighborhood preservation. (b) Silhouette score measures label separation.

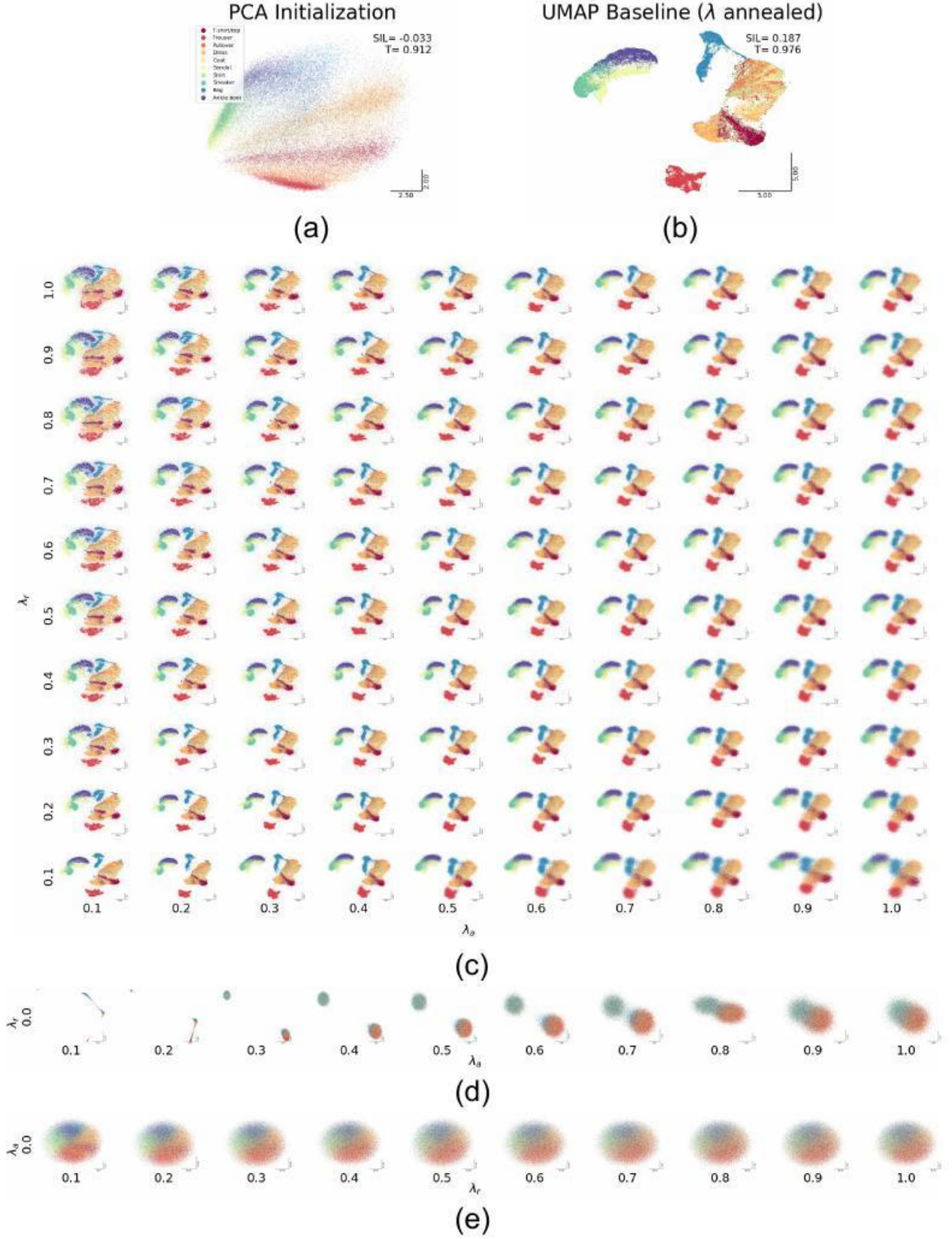


Figure 24: FMNIST sweep of λ_a and λ_r using UMAP attraction and repulsion shapes. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

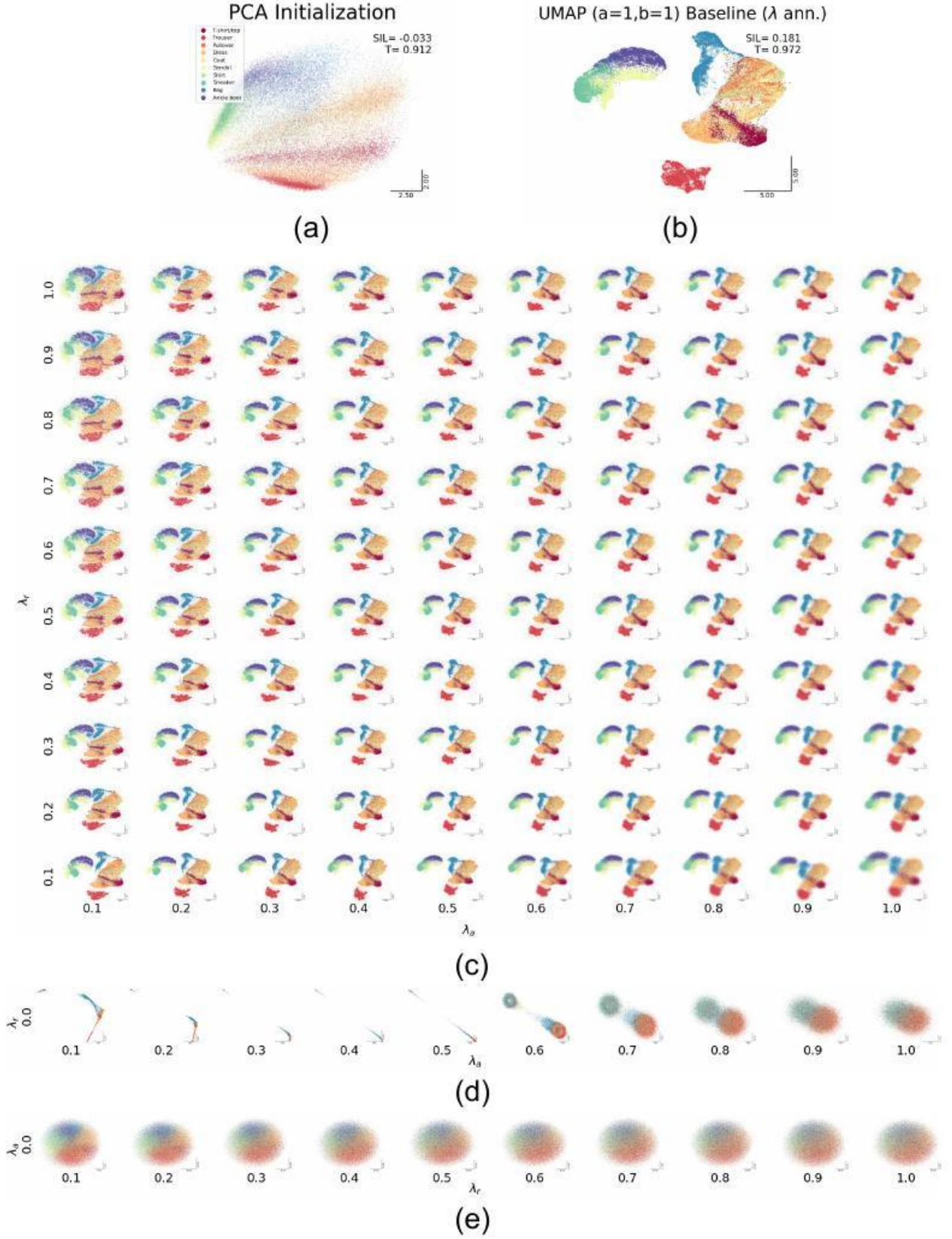


Figure 25: FMNIST sweep of λ_a and λ_r using UMAP shapes with $a = 1$ and $b = 1$. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

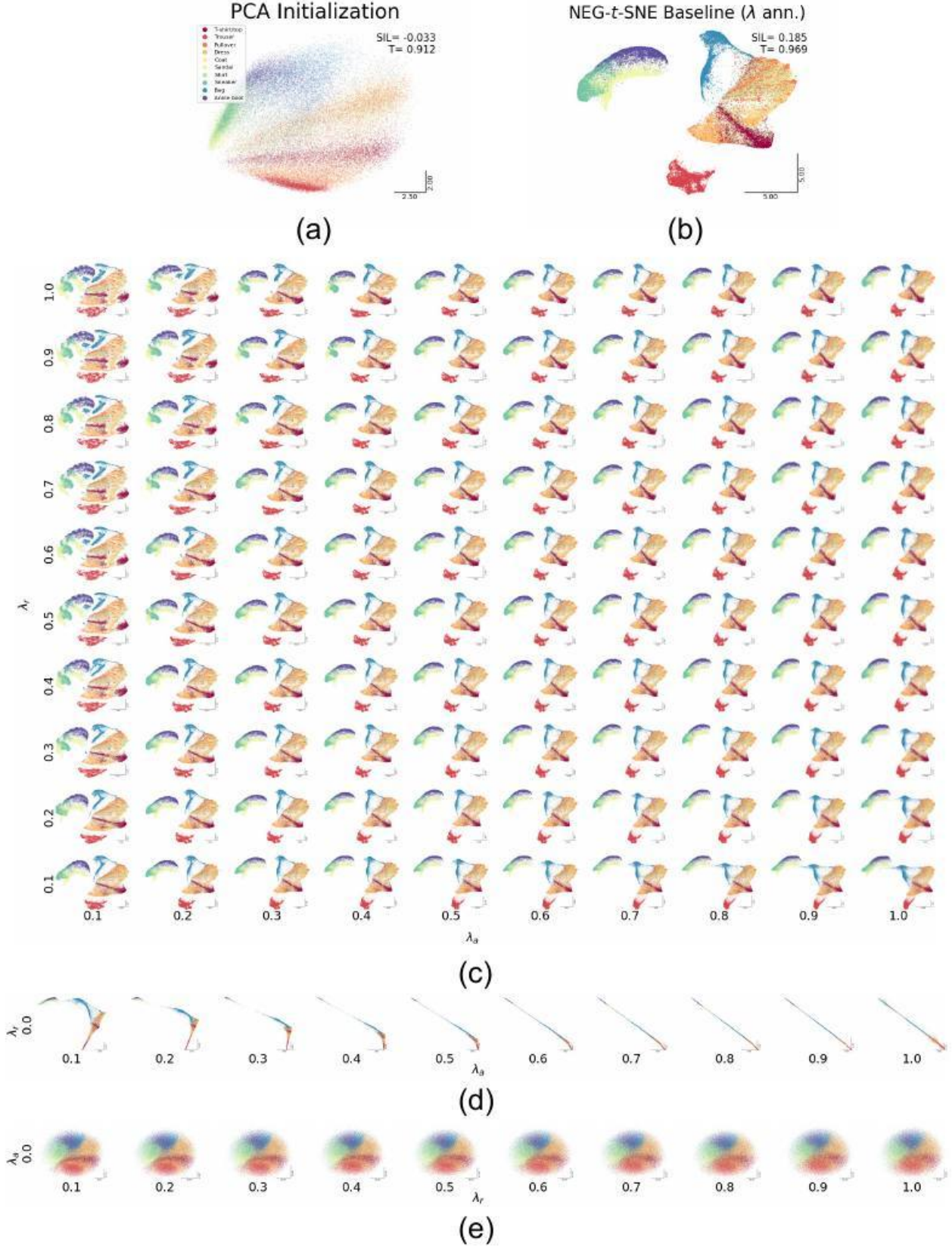


Figure 26: FMNIST sweep of λ_a and λ_r using NEG-t-SNE attraction and repulsion shapes. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

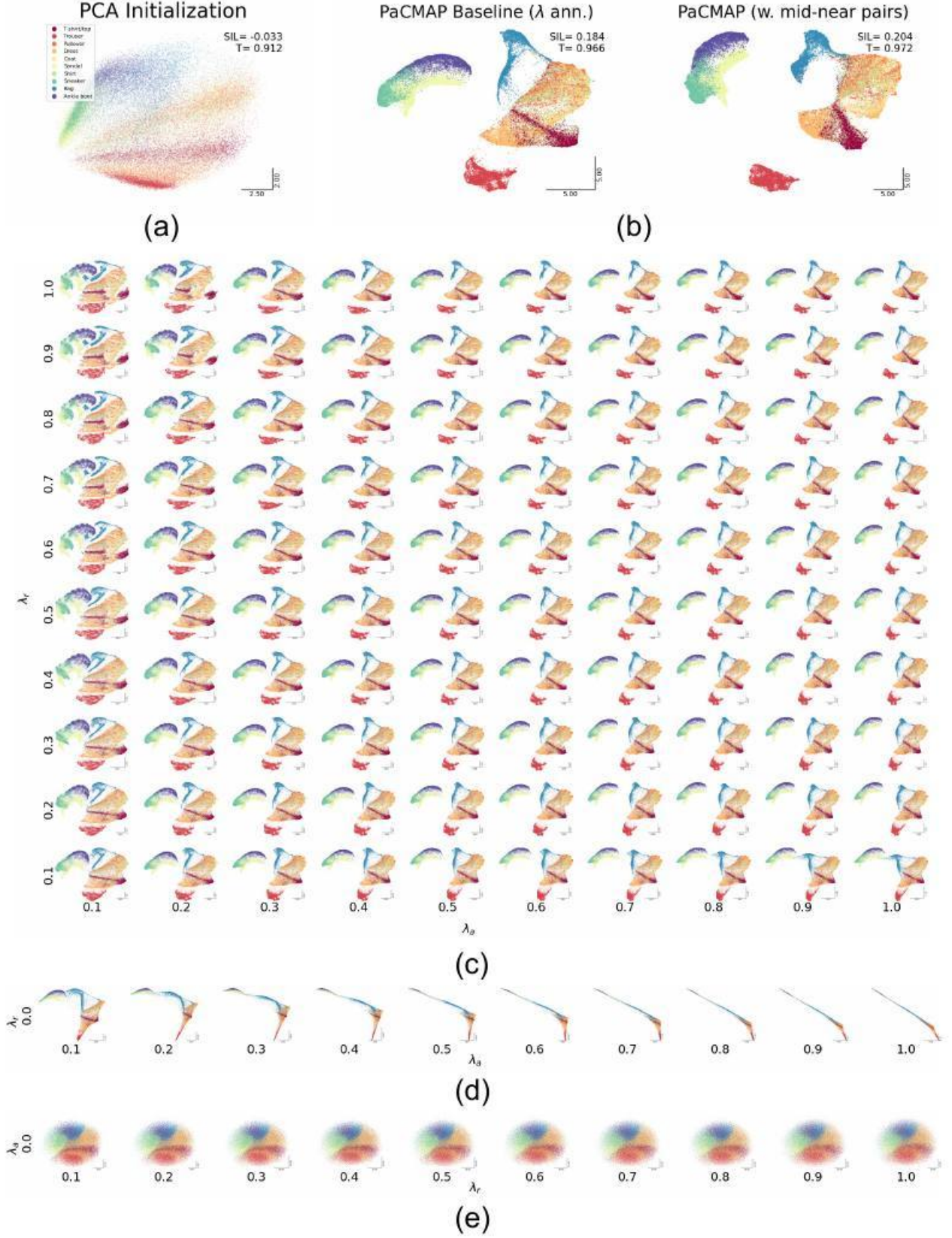


Figure 27: FMNIST sweep of λ_a and λ_r using PaCMAP attraction and repulsion shapes. (a) Shared initialization. (b) Left: annealed- λ baseline starting from 1 with mid-near points excluded to isolate attraction-repulsion interaction; right: reference with mid-near points included. (c) Constant-rate embeddings across the λ_a , λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

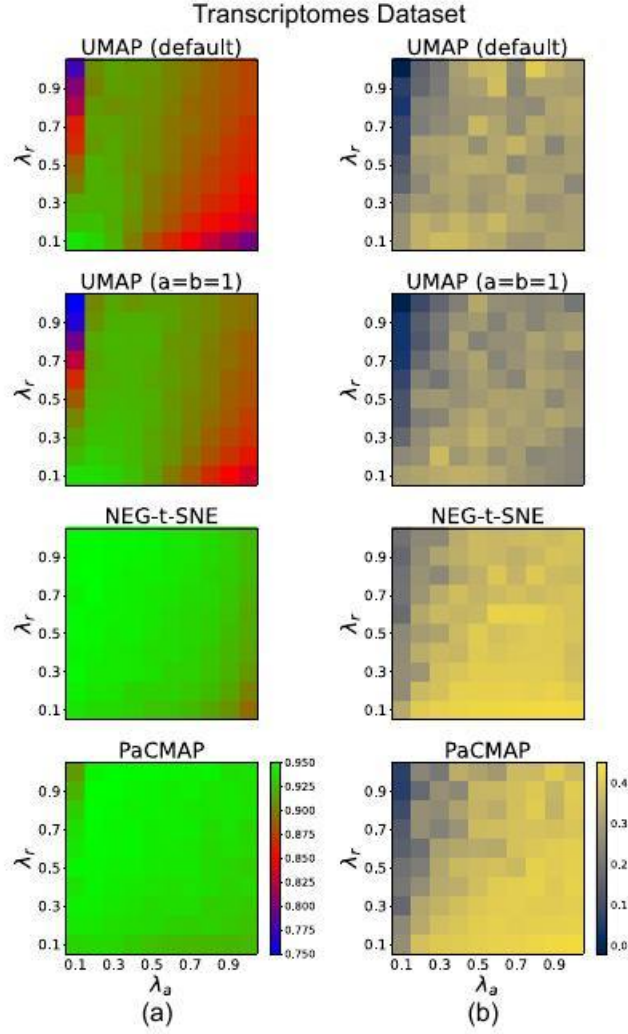


Figure 28: Single-cell transcriptomes metric comparison across methods. (a) Trustworthiness measures neighborhood preservation. (b) Silhouette score measures label separation.

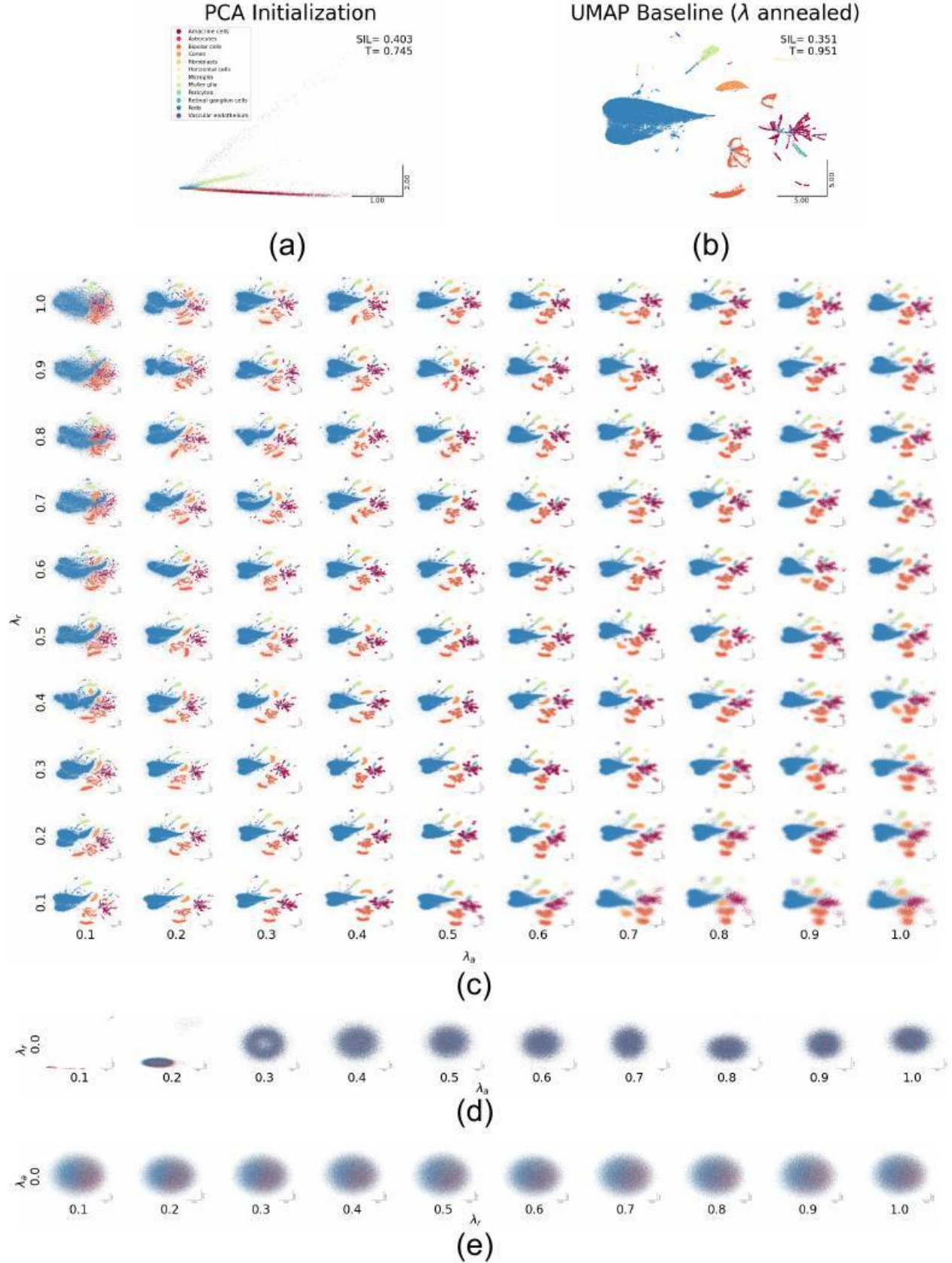


Figure 29: Single-cell transcriptomes sweep of λ_a and λ_r using UMAP attraction and repulsion shapes. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a , λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

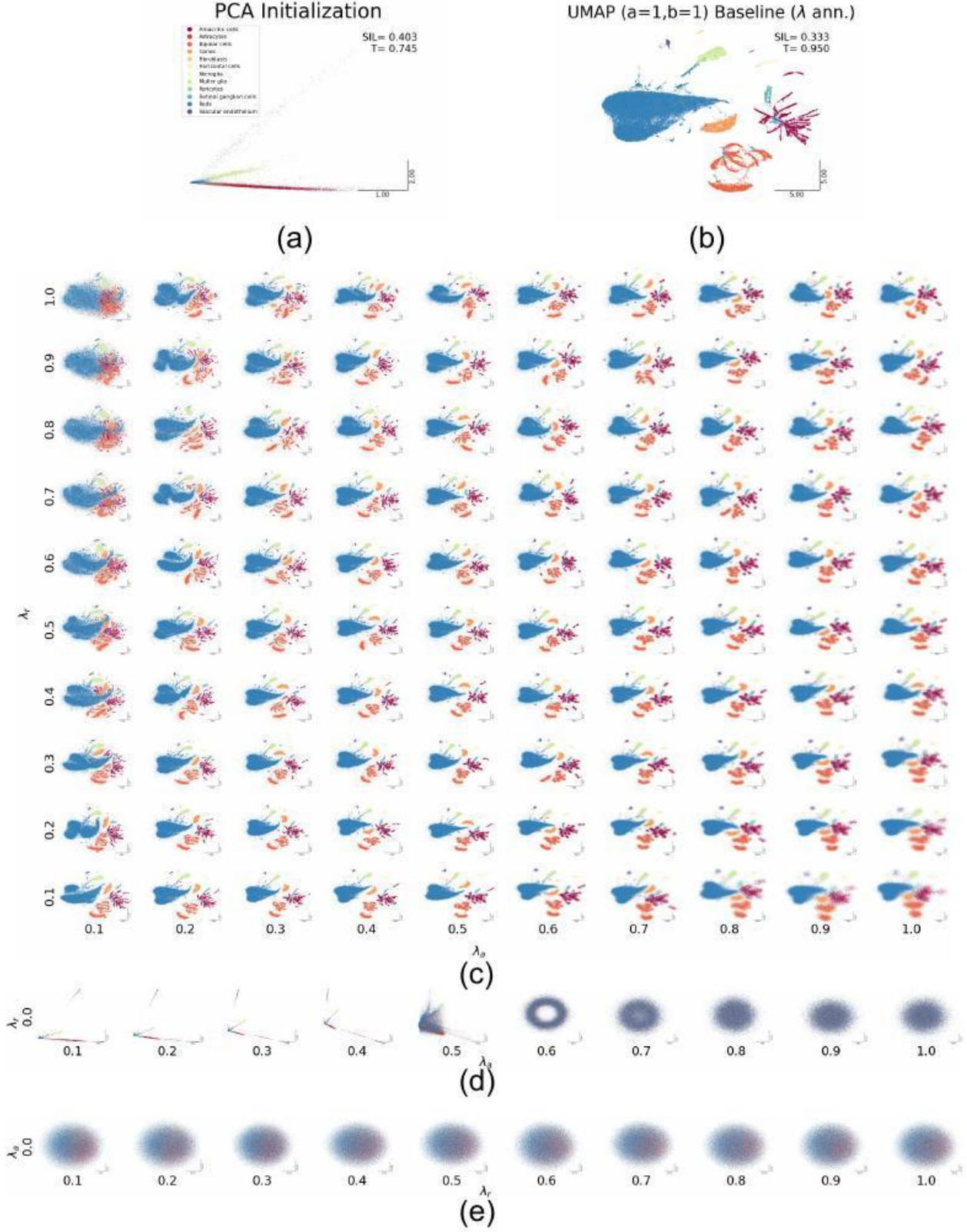


Figure 30: Single-cell transcriptomes sweep of λ_a and λ_r using UMAP shapes with $a = 1$ and $b = 1$. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a , λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

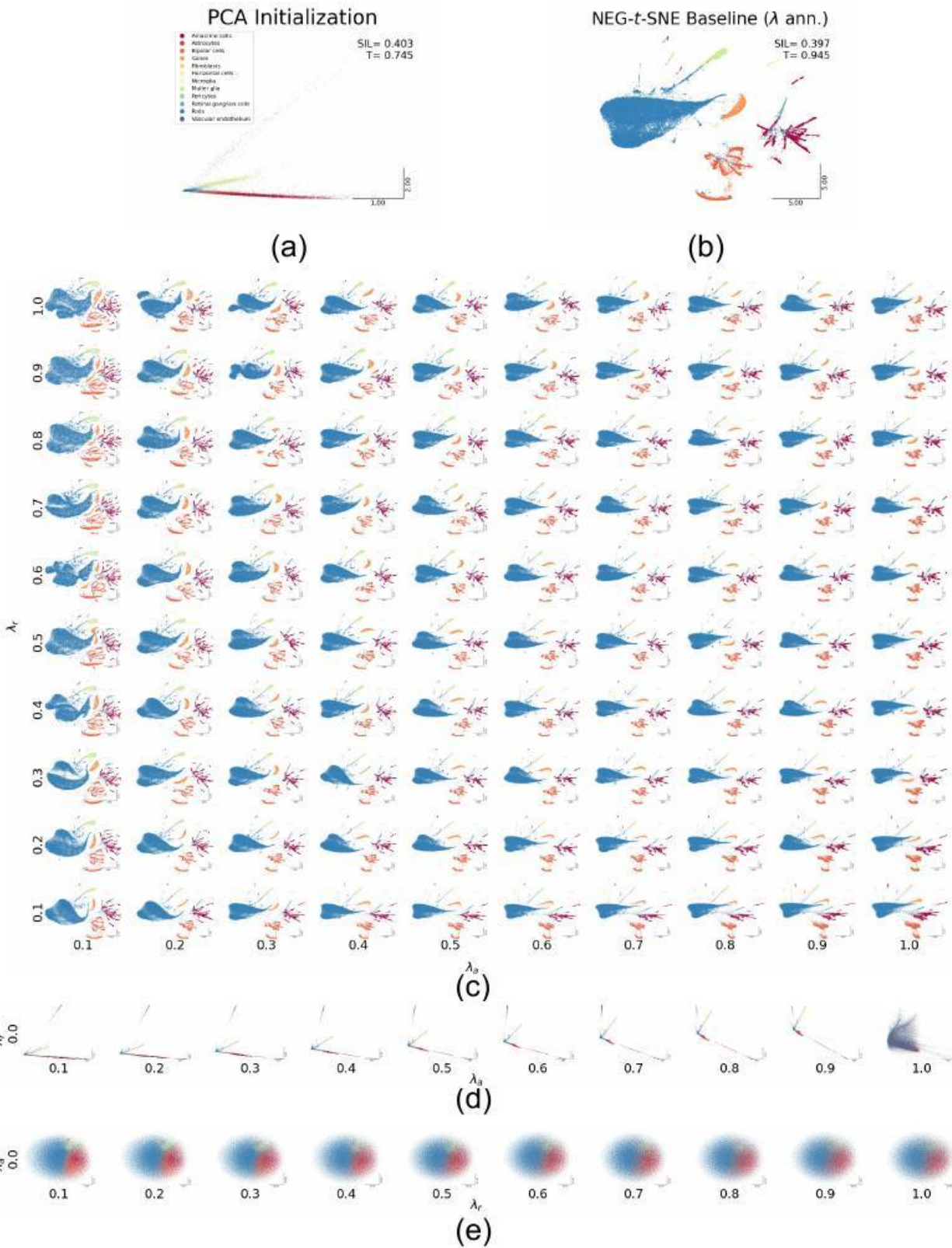


Figure 31: Single-cell transcriptomes sweep of λ_a and λ_r using NEG-*t*-SNE attraction and repulsion shapes. (a) Shared initialization. (b) Annealed- λ baseline starting from 1. (c) Constant-rate embeddings across the λ_a, λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not produce distinct clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

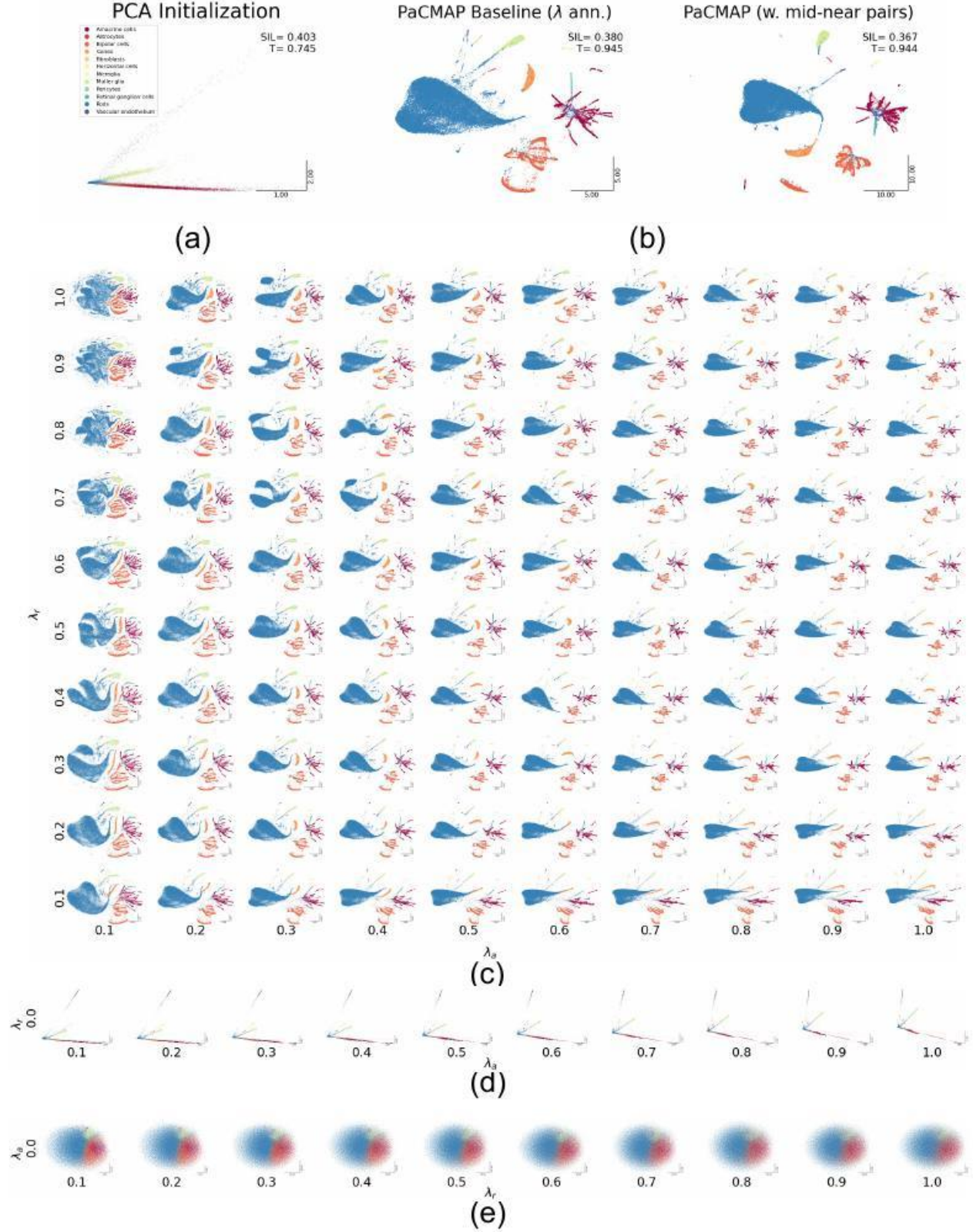


Figure 32: Single-cell transcriptomes sweep of λ_a and λ_r using PaCMAP attraction and repulsion shapes. (a) Shared initialization. (b) Left: annealed- λ baseline starting from 1 with mid-near points excluded to isolate attraction-repulsion interaction; right: reference with mid-near points included. (c) Constant-rate embeddings across the λ_a , λ_r grid, with no annealing. (d) With $\lambda_r = 0$, attraction alone does not form clusters. (e) With $\lambda_a = 0$, repulsion alone does not form clusters.

J Mix and Match Approach

Our shape analysis treats attraction and repulsion as separable design components, which suggests a controlled mix-and-match experiment. Figure 33 combines attraction and repulsion shapes derived from different losses while holding the optimizer and constant learning rate fixed at $\lambda = 1$. To keep the comparison within the contraction guarantee of Proposition 4.1 the attraction shapes are restricted to $[-1, 0]$; to avoid extreme short-distance repulsion, the repulsion shapes are finite. Under this common evaluation condition, UMAP’s scaled attraction shape ($f_a = 0.5f_a^U$) gives better clustering as reflected by higher silhouette score, whereas PaCMAP’s attraction shape gives better local structure as reflected by higher trustworthiness. The latter advantage is consistent with PaCMAP’s attraction lying within $[-0.5, 0]$, which prevents flips during updates. When UMAP and NEG- t -SNE attraction shapes are also confined to $[-0.5, 0]$ (using $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively), local structure improves at the expense of clustering performance (Fig. 34).

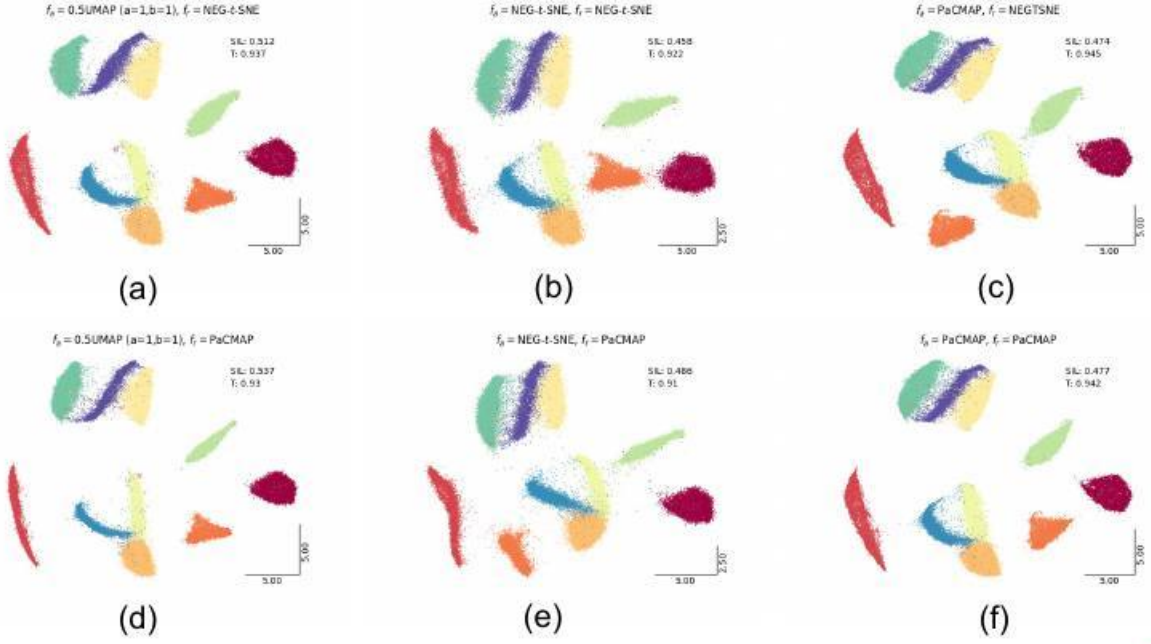


Figure 33: MNIST mix-and-match comparison at constant learning rate $\lambda = 1$. Attraction shapes satisfy Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with NEG- t -SNE repulsion. Bottom row: the same attraction shapes paired with PaCMAP repulsion.

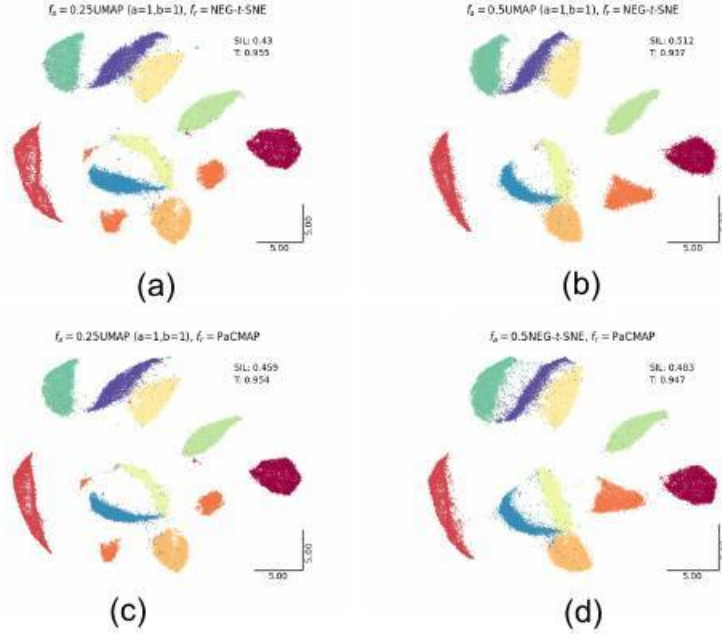


Figure 34: MNIST mix-and-match comparison with attraction confined to $[-0.5, 0]$ at constant learning rate $\lambda = 1$. This stricter attraction margin prevents flips during attractive updates. Top row: UMAP and NEG- t -SNE attraction shapes paired with NEG- t -SNE repulsion. Bottom row: the same attraction shapes paired with PaCMAP repulsion.

K Implementation Details

For the controlled comparisons, we implemented our own UMAP algorithm. We used `numba` (Lam et al. 2015) to compute an exact nearest-neighbor graph, rather than an approximate graph, with $k = 15$, and used `scikit-learn`'s (Pedregosa et al. 2011) PCA implementation for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1 2 3 4 6 8 9 and 10. Unless otherwise stated, the same implementation was used when replacing UMAP attraction and repulsion shapes with those of alternative methods (Figs. 18 34). Trustworthiness and silhouette scores were computed with the corresponding `scikit-learn` functions.

For visual comparability, mappings in Figs. 2 3 9 10 and 15 are rotated to the corresponding reference embedding in panel (a). We perform this Procrustes alignment by normalizing embeddings to zero mean and unit norm and then using SciPy's (Virtanen et al. 2020) `orthogonal_procrustes` method to estimate rotation and scaling.

For the UMAP versus NEG-*t*-SNE comparison in Fig. 11 we used the original contrastive-embedding implementation for both algorithms; implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15 16 22 (b) (right), 27 (b) (right), and 32 (b) (right) were generated with the official PaCMAP package; implementation details will be released upon acceptance.

The codes for displaying the embeddings of the PubMed dataset in Fig. 7 will be released upon acceptance.